

Highlights

Evaluation geometry and the limits of cross-region transfer in pre-fire thermal wildfire prediction

Yunus Emre Cogurcu, Emrehan Metin

- Scoring the same model at the fire, not region-wide, costs 0.143 AUC
- The cause is the negative pool, not prevalence: +0.155 against -0.000
- Under blocked CV, thermal predictors add +0.06 to +0.15 AUC in five regions
- Equalising the evaluation frame lifts mean transfer from 0.540 to 0.617
- None of 20 transferability diagnostics is shown to order transfer skill

Evaluation geometry and the limits of cross-region transfer in pre-fire thermal wildfire prediction

Yunus Emre Cogurcu^{a,*}, Emrehan Metin^a

^a*Çukurova University, Adana, Türkiye*

Abstract

Pre-fire thermal dryness separates a fire year from a normal year, but models built on it are rarely tested outside the region where they were fitted. Five Mediterranean wildfire regions were analysed on about 500 m cells, with MCD64A1 labels and spatially blocked validation, adding six thermal predictors to a terrain, fuel and greenness baseline. The first result is about **evaluation geometry**: holding the model, its predictors and its fitting fixed and changing only which cells are scored, moving from the whole region to the burn scar and its 2 km collar costs **0.143 ROC-AUC** [**+0.077, +0.208**], which is the size of the predictor-block increments this literature reports as findings. A control isolates the cause as the composition of the negative pool rather than class balance: matching prevalence costs -0.000 [$-0.003, +0.002$] while substituting fire-adjacent negatives costs $+0.155$ [$+0.093, +0.217$] over nine held-out scars. Applied between regions the same effect withdrew five of our own claims. The five study areas enclose very unequal far fields, and restricting each to a 10 km collar, which removes only far-field negatives, lifts mean transfer from 0.540 to 0.617 and leaves no sign reversal supported under

*Corresponding author.

Email address: `ycogurcu@cu.edu.tr` (Yunus Emre Cogurcu)

our own criterion, though the LST anomaly still differs between regions on a difference interval. The agreed direction is the opposite of the one dryness physics predicts: at the point estimates hotter pre-fire surfaces burned less in every region, and on mutual adjustment temperature survives where greenness does not, so the absolute channels behave here as static land-surface descriptors. Local skill nonetheless does not travel. The thermal block adds +0.056 to +0.153 within regions at 1 km blocking, +0.045 to +0.148 at 5 km, but +0.022 $[-0.032, +0.077]$ when a whole burn scar is withheld and +0.004 $[-0.028, +0.036]$ across twenty transfer directions, with a sign that varies by pair; the static baseline transfers no better, at 0.537 against 0.541 as drawn and 0.593 against 0.617 equalised; and equalised transfer of 0.617 sits against a matched within-region reference of 0.772, a shortfall of 0.155. Of twenty candidate diagnostics from five families, none was shown to order transfer: eighteen failed on the frames as drawn, and the two that succeeded require target labels and stop ordering transfer once the frames are equalised, one losing its variance entirely and the other its correlation (ρ +0.81 to -0.06). Transfer skill has to be measured, not inferred from similarity.

Keywords: wildfire, model transferability, area of applicability, land surface temperature, domain adaptation, concept shift

1. Introduction

Wildfire is a defining disturbance of Mediterranean-basin landscapes, and a changing climate is reshaping where and how it burns [29]. The literature on fire susceptibility mapping is large and has recently been synthesised [42, 14].

5 Its dominant pattern is settled: geospatial predictors are assembled over a

study region, paired with a historical record of burned area, and fitted with a supervised classifier, most often a random forest [5]; the resulting surface is published with a cross-validated AUC of 0.85 to 0.95.

That pattern does not report portability. Predictors describing the *state*
10 of a surface in one season are more informative about that season than predictors describing a place, and they do not travel. Establishing that is harder than it looks: the reading we first reached, that the relationship is reparameterised locally and points in different directions in different places, does not survive a correction to how the study regions were evaluated (Section 4.4).
15 What survives is that the dynamic block is worth +0.056 to +0.153 ROC-AUC inside every region under blocked cross-validation, +0.022 when a whole burn scar is withheld, and +0.004 between them on an interval spanning zero, with a sign that belongs to the source-target pair rather than to the block.

The obvious reading of that pair is a trade-off, and our earlier framing
20 took it. The data do not support it. The static baseline transfers at 0.537 against the thermal model’s 0.541, so the predictor class that ought to travel does not travel either, and removing the reversing predictors returns +0.014 on an interval that also spans zero. Two nulls on the portability axis are not an exchange. What is measured is a local gain that does not export and a
25 local cost when those predictors are removed, with no compensating transfer gain on either arm.

1.1. Portability goes unmeasured

A susceptibility model’s reported skill is almost always an estimate of *within-region* performance. Held-out folds come from the same study area
30 and season, often the same fire, and random folds over cells let spatial

autocorrelation inflate the estimate further. The problem is documented across ecological modelling [33, 31] and addressed by spatially blocked cross-validation [39, 25]. Blocking corrects what it corrects, but it does not make the estimate honest about a fire the model has not seen; Section 4.3 measures
35 that gap on identical cells.

Because only that side of the ledger is reported, portability is never entered. A predictor block is adopted on the increment it delivers inside its training footprint, and whether that increment survives a change of region is not asked, even though any regional product built from locally trained models implicitly promises generalisation beyond it. Meteorological fire-danger
40 indices are known not to port cleanly between fire environments [32], and the two recent studies that test model transfer systematically, across Mediterranean countries [8] and US counties [16], both report that it largely succeeds between similar regions — and both transfer models whose dominant predictors are *spatially stationary*. Whether a dynamic, season-specific predictor
45 class behaves the same way is the question this paper puts.

1.2. Pre-fire thermal dryness is the natural test case

Pre-fire thermal dryness is the dynamic class most plausibly *expected* to transfer. Standard susceptibility predictors are static or near-static over the
50 timescale at which fire danger varies, so they explain poorly why one summer burned and the preceding one did not. What changes between those summers is the state of the surface, to which satellite thermal observation gives partial access (Section 2.2). The Temperature-Vegetation Dryness Index [35] formalises that access into an internally normalised measure, which should
55 in principle be less exposed to absolute-temperature offsets between regions

than raw surface temperature. **That theoretical portability advantage is tested here and is not found:** the two normalised channels transfer no better than the four absolute ones (Appendix A(f)).

The physics linking moisture stress to combustion is universal, so portability should be most expected for this class, which is what makes it diagnostic: a loss here cannot be dismissed as a peculiarity of a locally defined covariate. That expectation motivates the design; it is not a finding of it. Section 4.4 reports that the signed associations run the other way — a hotter pre-fire surface is associated with *less* burning in all five regions, and on mutual adjustment temperature survives where greenness does not, so on this cohort the absolute channels behave as static land-surface descriptors rather than as a dryness index. We keep the framing because it is why these predictors were chosen and why their failure to travel is informative, and we state the

1.3. Why the loss is invisible to the diagnostics in use

A deficit on transfer can arise two ways. Under **covariate shift** the marginal distribution of the predictors differs while the predictor-response relationship is preserved, which is in principle correctable without target labels, by per-region standardisation or covariance alignment such as CORAL [37], an approach with an established remote-sensing literature [38]. Under **concept shift** the conditional relationship itself changes: a predictor positively associated with burning in one region is negatively associated with it in another. No label-free realignment of input distributions can repair a sign reversal, because the information needed to detect it is exactly the information being withheld. The taxonomy is canonical [27].

This distinction is what makes the failure hard to anticipate. The di-

agnostics the field relies on to decide whether a model may be applied to new ground, above all area-of-applicability measures built on dissimilarity in predictor space [23, 24, 18], are marginal by construction. A target region can sit well inside the training data’s predictor envelope while the relationship between those predictors and burning points the other way. This paper
85 measures both sides on the same pairs.

1.4. Contributions

Three findings carry this paper, stated here as claims and established in Section 4, which carries every interval.

90 **Contribution 1. Where a model is scored decides what it appears to know, and the effect is large enough to dissolve findings of our own.** That evaluation extent inflates AUC is established in species distribution modelling (Section 2.4); what is new is a magnitude on the wildfire problem under a controlled design. Holding the model, the predictors and
95 the fitting fixed and changing only which cells are scored costs **0.143 ROC-AUC**, the size of the predictor-block increments this literature publishes as findings, with the cause isolated to the composition of the negative pool rather than class balance (Section 4.3). Applied between regions the same effect withdraws five claims of our own, including the sign reversal we had
100 offered as the mechanism of the transfer residual and the one arm that held place fixed (Section 4.4). The practical consequence is a reporting standard (Section 5.8).

Contribution 2. Local skill does not travel, and correcting the frame does not rescue it. The thermal block is worth a substantial within-
105 region increment in every one of five regions under blocked cross-validation,

with every bootstrap interval above zero, and it remains positive in all five when the frame is equalised. Much of that is a property of interleaved hold-out: withholding a whole burn scar leaves an increment whose interval spans zero. Across twenty ordered transfer directions its paired contribution also
110 spans zero, with a sign that is a property of the pair rather than of the block. Two controls bound the reading. The static baseline transfers no better than the dynamic one on either frame, so the failure is not the thermal block’s peculiarity; and on matched frames and matched blocking the equalised transfer still falls **0.155** short of the within-region reference. The
115 within-region half is not novel [1, 13]; the paired contrast against portability is. Dimarco et al. [8] transfer a predominantly stationary predictor set successfully across a comparable Mediterranean design, though their response variable is ignition rather than burned area (Section 5.7).

Contribution 3. The shortfall cannot be anticipated by any
120 **diagnostic we could run.** Twenty candidate diagnostics from five families are evaluated against observed transfer under one bootstrap framework, including predictor-space dissimilarity [23, 24, 18], climatic and geographic distance, learned domain separability, niche overlap and burn-pattern regime distances. Eighteen were not shown to order the matrix on the frames as
125 drawn. Two conditional variants did, but both need burned labels on both sides, both rest on a data-selected feature subset whose all-nine counterparts span zero, and **Contribution 1 removes even those:** on comparable frames one becomes unanimous with no variance left to correlate and the other loses its correlation entirely (Section 4.4). The point survives without
130 any ranking: the pair with the highest burned-niche overlap is among the

weakest in the matrix while the lowest transfers above chance in both directions, contradicting an expectation the fire literature carries [8, 16] and agreeing with species distribution modelling [41, 34].

Two consequences follow, reported as supporting results. Label-free alignment by standardisation and covariance alignment [37] does not repair transfer, and we believe this is the first application of covariance alignment to fire susceptibility. And because the residual is conditional, the resource that closes it is target labels, whose price is measured in Section 4.12.

Alongside these, a leakage-audited, spatially blocked evaluation and transfer protocol is released with code, configuration and frozen outputs, so that most of this result can be re-run rather than taken on trust. One component is an exception and is named in the data-and-code availability statement: the reproduction-check code is not in the released repository at the commit of record. The few-shot export, previously also listed as an exception, was regenerated at commit `6d7a6a71` and its validator now passes on every check. This matters given evidence that wildfire transfer conclusions are sensitive to evaluation design and task formulation [45].

A companion paper treats the observational layer beneath this one, where preprocessing decisions taken before any model is fitted are shown to carry their own uncertainty budget.

Section 2 reviews the relevant literature. Section 3 describes the study regions, the data, and the modelling, cross-validation, transfer and adaptation protocol. Section 4 reports results. Section 5 discusses them, including limitations. Section 6 concludes.

155 2. Related work

2.1. Fire susceptibility mapping, and what it reports

The dominant pattern is described in Section 1; the figure it publishes, a cross-validated AUC of 0.85 to 0.95, is a within-region estimate. Comparable within-region results already exist for the landscapes studied here [1, 13].

160 *2.2. Pre-fire thermal dryness*

Satellite thermal observation gives repeated access to surface state through fuel moisture content [47]. Chuvieco et al. [7] established the pairing of land surface temperature with a vegetation index as a live-fuel-moisture estimator for fire-danger rating, and the Temperature-Vegetation Dryness Index [35] formalises that feature space into an internally normalised measure, which is
165 why it is often expected to travel between regions better than raw temperature. That pre-fire thermal state carries information about subsequent fire is established [19, 21, 20]. Gelabert et al. [9] found dead fine fuel moisture and its anomalies the most influential predictor of human-caused ignition likelihood across Europe, testing generalisation by pooled fitting with per-site
170 evaluation. What has not been tested is whether a classifier built on this class survives strict, label-free application to a region it never saw.

2.3. Spatial validation and transferability

Random cross-validation over spatially autocorrelated cells inflates skill
175 estimates [33, 31], and spatially blocked designs are the standard remedy [39, 25], though blocked schemes have themselves been argued to introduce pessimistic bias [43, 26, 6]. This paper blocks throughout and reports every

result at three block sizes, so a reader can see which verdicts depend on the choice.

180 Blocking corrects the inflation random folds produce within a region. It does not speak to performance on a fire the model has not seen, which Section 4.3 measures on identical cells. For that, the field’s instrument is the area of applicability and related predictor-space dissimilarity measures [23, 24, 18], which ask whether a target’s predictor values fall inside the
185 training data’s envelope. Species distribution modelling has examined transferability far more systematically than fire has [46], and two studies there report that geographic and environmental similarity do not reliably predict transfer success [41, 34]; transferability of satellite-derived empirical models is an active question in fire remote sensing itself [22].

190 The canonical taxonomy separates covariate shift, where the predictor distribution moves but the predictor-response relationship holds, from concept shift, where the relationship itself changes [27]. Covariate shift is in principle correctable without target labels, by per-region standardisation or by covariance alignment such as CORAL [37], and domain adaptation has
195 an established remote-sensing literature [38, 30]. Concept shift is not: no realignment of input distributions can repair a reversal in the sign of an association, because detecting it requires the labels being withheld. Shift decomposition in applied remote sensing is not itself new [12]. We are not aware of a prior application of covariance alignment to fire susceptibility, fire
200 occurrence or burned-area prediction.

2.4. Cross-region generalisation of fire models

Few studies test fire-model transfer directly. Podschwit et al. [32] report, in a Peruvian case study, that meteorologically derived danger indices do not port cleanly between fire environments. WildfireGenome [16] runs
205 a leave-one-county-out matrix across seven US counties and reports strong within-county performance with highly variable off-diagonal transfer, similar pairs transferring well and dissimilar pairs collapsing; its label is a principal-component composite of hazard *indicators* rather than observed burned area, and it applies no adaptation. Xu et al. [45] argue that wildfire transfer con-
210 clusions depend strongly on evaluation design, a caution we address by fixing the protocol in a project log before the diagnostics were computed and by reporting every sensitivity axis; that log is not a formal pre-registration, and Section 3.11 states what was fixed and when. Kondylatos et al. [15] provide Mesogeos, a 1 km Mediterranean datacube.

215 **Evaluation extent and AUC.** Species distribution modelling settled long ago that the area a model is evaluated over is not a neutral choice. Lobo et al. [17] make it the fifth and, by their own ranking, most important reason to distrust AUC as a comparative measure: the extent of the modelled area governs how many easy absences enter the calculation, and therefore the
220 score. VanDerWal et al. [40] show the same lever on the calibration side, and Barve et al. [3] give the argument its general form as the accessible area. That literature is qualitative about magnitude, because magnitude is problem-specific. Two things follow here. Our Contribution 1 is a measurement inside that framework rather than a new phenomenon, and Section 1 states
225 it that way. And the wildfire literature has largely not imported the lesson:

region-wide figures are reported as though they described performance at the fire, and we know of no wildfire study that holds the model fixed and varies only the evaluation cells. The 0.143 we obtain is the size of the predictor-block increments this literature reports, which is why Section 5.8 treats it as
230 a reporting problem rather than a caveat.

The nearest neighbour, and the contrast the present paper draws. Dimarco et al. [8] is the closest Mediterranean analogue: 500 m predictors harmonised across four countries, tree ensembles under spatial cross-validation, transfer tested both leave-one-country-out and as a full 4 x 4
235 matrix, with every transfer exceeding AUC 0.80 and bioclimatically similar countries scoring higher. They apply no domain adaptation.¹

Much of the design is shared — Mediterranean regions, 500 m cells, MCD64A1-derived targets, tree ensembles, spatially aware validation, an explicit transfer matrix — and two things differ. **The predictor class:** their
240 model rests on attributes of a place, all spatially stationary, ours on the state of a surface in one season. **The response variable:** theirs is human-driven ignition, ours is burned area, and those are different quantities with different dominant controls. Read together the two results suggest that the relationship between domain similarity and transfer success may be predictor-class
245 dependent, with Vesk et al. [41] and Rousseau and Betts [34] aligning from species distribution modelling while Dimarco et al. and WildfireGenome run against it. We present that as a live disagreement, and Section 5.7 states why our own data do not settle it.

¹Their Results text refers to "LST anomalies", which is inconsistent with their own

Methods, where no land surface temperature or TVDI variable appears
 250 and the only temperature predictor is a static ERA5-Land seasonal climatology [28]. We follow their Methods, which we take to be authoritative, and note the discrepancy so that the comparison is transparent. Every statement about their work was checked against the full text, read 2026-08-08.

3. Data and methods

255 3.1. Study regions and temporal windows

Five Mediterranean wildfire regions are analysed (Fig. C.1): Manavgat 2021 and Muğla 2021 in Türkiye, Bejís 2022 in Spain, North Evia 2021 in Greece and Montiferru 2021 in Sardinia. Each is a place-based rectangular area of interest in EPSG:4326, defined from place coverage rather than from
 260 a fire perimeter, deliberately not clipped to it so that unburned cells around each fire form the negative class, and not tuned on burned prevalence, gate outcome or any model metric. **One AOI choice was label-informed and is stated as such:** the North Evia box was extended after the legacy box was found to carry an atypically high burned prevalence, the extended geometry
 265 then being defined from place anchors, with the legacy variant kept as a sensitivity arm (Appendix A). Section 4.4 shows that this framing decision is consequential and Section 5.9(x) treats it as the design lesson of the paper. A sixth region, Kozan 2023, is carried as a negative control and excluded by the gate of Section 3.3.

270 Each region has two non-overlapping windows. The **predictor window** closes the day before the **label window** opens, so no predictor observation can post-date the first labelled burning. Window lengths vary between re-

gions with the event: 57 to 61 days for predictors and 35 to 59 days for labels, counted inclusively. A four-year baseline of window-symmetric composites
 275 precedes each predictor window and supplies the climatological reference for the anomaly channels. Region identifiers, bounding boxes, window dates and baseline years are registered in `core/regions.py` and are reproduced here.

Region identifiers, bounding boxes, window dates and baseline years are registered in `core/regions.py` and reproduced in Appendix C, Table C1.

280 3.2. Burned-area label and the ~ 500 m analysis grid

Labels come from MODIS MCD64A1 Collection 6.1 [10] retrieved through Google Earth Engine [11], whose omission and commission characteristics [4] bound every model here. The analysis grid is **reconstructed rather than native**: the pipeline’s 30 m EPSG:4326 reference grid is partitioned into
 285 17 x 17 blocks, giving a nominal 510 m cell that approximates rather than reproduces the MODIS cell, and is square in degrees but not on the ground. A cell’s burn date is the mode of its positive sub-pixel day-of-year values, tested against the region’s label window; the label never affects a cell’s eligibility for modelling. Two safeguards are recorded rather than assumed: cells burning
 290 inside a region’s own predictor window are removed, which ran for three of five regions, and burning in earlier years is screened for none. Appendix C.1 gives the full specification, including what follows from the grid’s shape.

3.3. Burned-landcover admissibility gate

Before any modelling each region passes a gate asking what fraction of
 295 its burned cells is dominated by natural vegetation, using ESA WorldCover classes on the same cells. A region is admitted at 0.50 with at least 30

burned cells, and rejected as a cropland-dominated control at 0.50 cropland. The purpose is to separate burned area produced by natural-fuel combustion from post-harvest stubble burning, which MCD64A1 does not distinguish. Verdicts are in Section 4.1; the full rule is in Appendix C.1.

3.4. *Predictor variables*

Two feature sets are used. The **baseline** is terrain, fuel and greenness: elevation, slope, dominant land cover and a predictor-window median vegetation-index composite, all static or near-static over the timescale at which fire danger varies. The **thermal** set adds six pre-fire channels: current land surface temperature, its anomaly against a four-year window-symmetric baseline at the same cell, the Temperature-Vegetation Dryness Index and its difference against that baseline, and two coordinate-informed products, a downscaled and a fused surface temperature. The two internally differenced channels are the ones constructed to isolate the dynamic anomaly. TVDI's wet and dry edges are percentiles of the values a given area and window happen to contain, so it is not portable as a physical quantity independently of any concept shift; Section 5.9 carries the consequence. Provenance, compositing and the downscaler are in Appendix C.

3.5. *Cell aggregation, validity and analysis populations*

Cell-level values are means over the ~ 510 m cell, computed from valid 30 m pixels only, with the valid fraction recorded. A cell is **valid_for_modeling** when it meets a 30 % valid-pixel floor and no thermal channel is missing. Two populations are then defined. The **primary** population is natural vegetation, cells whose combined tree, shrub and grass fraction

reaches 0.50, which excludes cropland from every burnable mask. The **secondary** population is all valid cells, reported throughout as a sensitivity. Every headline result is given for both.

3.6. *Feature sets and classifier*

325 Two nested feature sets are compared. The **baseline** set is NDVI, elevation, slope and land cover. The **thermal** set is the baseline plus the six thermal channels. Land cover is one-hot encoded and numeric features are standardised inside the fitting pipeline. The classifier is a random forest [5] with 300 trees, unlimited depth, `min_samples_leaf = 3`, balanced
330 class weights and `random_state = 42`, identical for every region, population, feature set and transfer direction, so that no comparison in this paper is confounded by a model choice.

3.7. *Spatial-block cross-validation and bootstrap uncertainty*

Cross-validation is spatially blocked. Blocks are squares of the analysis
335 grid formed as `row_500m // B by col_500m // B`, and folds are drawn over whole blocks, so no block is split between training and test. Five folds are used. The block size B is reported at 2, 10 and 20 cells, about 1, 5 and 10 km, for the within-region analysis.

Uncertainty is a spatial-block bootstrap: blocks are resampled with re-
340 placement, 1000 replicates, seed 42, with 2.5 and 97.5 percentiles as the interval. Because it is blocks that are resampled, what bounds an interval's reliability is the number of blocks carrying at least one burned cell, and those counts are reported alongside the intervals. Where a verdict rests on too few such blocks it is stated as indicative rather than as an interval.

345 *3.8. Cross-region transfer protocol*

For each ordered pair of regions the model is fitted on the source population and applied to the target population with **no target label used at any point**: no refitting, no threshold selection, no calibration. Both the baseline and the thermal feature set are transferred, so that the thermal
350 block’s contribution is a paired difference within a direction rather than a comparison across directions. All twenty ordered directions among the five regions are computed.

3.9. Label-blind domain adaptation

Two label-free remedies are tested on every direction, in both feature sets.
355 **Region-wise z-score.** Each region’s numeric features are standardised using its own statistics, source statistics from source data and target statistics from target data, never pooled. The classifier is refitted on the z-scored source and applied to the z-scored target. This removes first- and second-order marginal offsets.

360 **CORAL after region-wise z-score.** Starting from the z-scored features, the source covariance is aligned to the target covariance by the standard whitening-recolouring map [37], with regularisation $\lambda = 10^{-5}$ and eigenvalues floored at 10^{-12} . Critically **the transform is applied to the source only**; the target is left exactly as it is. The classifier is refitted on the aligned
365 source. Neither variant sees a target label, and both are verified label-blind at run time.

λ sensitivity was assessed over nine values from 0 to 10^{-1} on four directions in both feature families, moving transfer AUC by at most 0.014 in

any direction and 0.008 within the thermal family. The canonical $\lambda = 1$ lies
 370 outside the released sweep and is reported separately in Appendix A(b).

3.10. *Transfer-gap decomposition and the concept-shift criterion*

The transfer gap is decomposed into the part a label-free covariate correction recovers and the part it does not, as (adapted – raw) / (within – raw), signed and unclipped, with its interval from the same paired bootstrap;
 375 Section 4.3 shows the remainder should not be read as a conditional residual, because much of it is incurred inside a single region (Appendix A(j)).

The mechanism is diagnosed by **signed univariate association**: for each numeric predictor the raw ROC-AUC against **burned** is computed in each region and never folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a value below 0.5 is a
 380 direction rather than weakness. A reversal is called bootstrap-supported only when the two regions’ point estimates fall on opposite sides of 0.5 **and each region’s own interval excludes 0.5**, under the same 10-cell spatial-block bootstrap. That is stricter than requiring the two intervals to be disjoint: a feature whose intervals are disjoint but one of which straddles 0.5 has not
 385 been shown to point anywhere in that region, so it is recorded as a point reversal only.

3.11. *Transferability diagnostics versus transfer*

Twenty candidate diagnostics from five families are computed for every region pair and rank-correlated (Spearman) against observed raw transfer
 390 AUC, under one bootstrap framework that resamples unordered region pairs with both of their ordered directions travelling together. The families are

marginal predictor-space measures $P(x)$, including area-of-applicability dissimilarity, climatic and geographic distance and a learned domain classifier; burned-niche overlap $P(x|y=1)$; burn-pattern regime distance $P(y)$; and conditional direction agreement $P(y|x)$. No family-wise error control is claimed, and the number of variants computed within each family is reported with the result. Appendix C.2 gives the full specification of each diagnostic and of the bootstrap.

3.12. Interventions

Pooled multi-region training. Leave-one-region-out: the model is trained on the pooled primary populations of the other four regions and evaluated on the held-out region, with folds blocked as before. This asks whether pooling recovers what single-source transfer loses.

Removal of direction-reversing features. The two predictors whose signed association reverses between regions with bootstrap support **on the frames as drawn** are **elevation_mean** (Manavgat against Bejís and against Muğla) and **lst_anomaly_mean** (Bejís against Evia). Section 4.4 withdraws that support under an equalised frame, so this selection rule is frame-dependent and the arm below is reported as a measurement made under the original protocol rather than as a consequence of an established reversal. They are dropped and everything is refitted, within-region and across every direction, and the two are also dropped singly so the cost can be attributed. This measures what the removal costs locally and what it returns on transfer on the same footing. Note that the two features are selected by the same reversal analysis against which the result is then read, so both quantities are post-selection estimates and no correction for that selection is applied.

3.13. Controls on the transfer path

Four evaluations establish what the transfer arms are measuring. All use the transfer protocol of Section 3.8 unchanged — fit on the source cells, apply
420 to the target cells, no refit, no recalibration, no threshold selection — and all were verified to reproduce the frozen cross-region AUCs before being used. They are a **within-region half-split** (the modelled cells cut in two at the median of a grid axis, both axes and both directions, a split discarded when either half is single-class); **leave-one-scar-out** (each burned component of
425 at least 50 cells, dilated by a 2 km buffer, withheld from training and used as the target); a **foreign-region evaluation** of the same held-out scar areas; and the **same blocked model restricted** to those areas, which isolates the evaluation region from the training regime. Scars are dilated by buffers of 2, 5 and 10 km, with 2 km reported as primary; the minimum-component rule,
430 and the per-split and per-scar positive counts, which are unequal and bear on the interpretation, are in Appendix A(i).

3.14. Leakage control and reproducibility

An explicit forbidden-column set is enforced at every model fit as an assertion rather than a convention: coordinates and their normalised forms,
435 every burn-date and label-provenance column, and the agreement fraction are excluded from all feature sets. The natural-vegetation mask defines the population and is never a predictor. All randomness uses seed 42 and the bootstrap 1000 replicates. Because the transfer analysis runs in an environment separate from the upstream pipeline’s, every within-region model
440 was refitted and compared against the frozen upstream output: the within-region comparisons agree exactly and the twenty transfer directions to within

1.6×10⁻⁷. Section 5.9(vi) states the implementation tolerance that applies if the library version is not pinned. Every headline result is repeated across two populations, three block sizes, the CORAL sweep, both feature sets and four
445 classifier capacities, and where a conclusion depends on one of those choices the dependence is reported rather than resolved by choosing the favourable setting (Appendix A). Full detail is in Appendix C.

3.15. *Same-geography event-to-event comparison (Muğla 2021 versus 2022)*

Muğla admits a comparison in which place is held fixed and the event
450 varies: a second fire burned inside the identical AOI, on the identical analysis grid, eleven months after the first. Signed univariate AUCs are computed for both arms under the same 10-cell spatial-block bootstrap used elsewhere. Season, year and population all differ between the arms, since the 2022 arm is defined by removing the 2021 scar, and Section 4.10 reports what that
455 costs. Appendix C.3 gives the window dates, the population construction and the asymmetry audit.

4. Results

4.1. *Study regions and the admissibility gate*

All five candidate regions pass the burned-landcover gate as wildfire can-
460 didates and the negative control fails it as intended. Kozan 2023 returns a natural-vegetation fraction of 0.017 against the 0.50 threshold and is excluded from all modelling. The separation is not marginal: the five admitted regions carry fractions of 0.723 to 0.991, so the threshold falls in an empty interval rather than between neighbouring cases. That is consistent with the

465 gate separating burned area produced by natural-fuel combustion from post-harvest stubble burning, which MCD64A1 does not distinguish, though one negative control cannot establish it.

North Evia is analysed on an extended AOI. Relative to the legacy box, the extended box leaves the burned scar essentially unchanged while cutting
 470 overall prevalence from 0.361 to 0.122 and TSG prevalence from 0.676 to 0.287. The effect of that choice on transfer is reported in Appendix A(a).

4.2. *Within-region: the thermal increment replicates in five regions*

Adding the six thermal predictors to the baseline raises spatially blocked out-of-fold ROC-AUC in every region. The increment’s bootstrap interval
 475 excludes zero in all five regions at 1 km and at 5 km blocking (Fig. C.3). Those are the two scales this design supports as intervals. At 10 km the point estimates hold, from +0.048 to +0.154. They rest on 6 to 33 positive-carrying blocks, so they are reported as indicative rather than as intervals. The table note gives the reason.

480 *Table note (resampling units).* The bootstrap resamples spatial blocks, so what bounds an interval’s reliability is the number of blocks carrying at least one burned cell, and those counts fall quickly as blocks coarsen: 235, 302, 843, 716 and 192 at 2 cells; 28, 19, 70, 41 and 16 at 10 cells; and **12, 6, 33, 15 and 6** at 20 cells, out of 60, 48, 167, 50 and 12 blocks. No bootstrap replicate
 485 was invalid at any block size except Bejís at 20 cells, where 5 of 1000 were single-class — the symptom of the six positive-carrying blocks just reported. An equal-tailed percentile interval built on six positive-carrying blocks has no meaningful coverage, and Montiferru at 20 cells feeds only 12 groups into a 5-fold grouped split, so its models train on about ten blocks each. **The**

Table 1: Within-region baseline versus thermal performance and block-size robustness. Primary (TSG) population; spatially blocked 5-fold CV (Section 3.7); paired spatial-block bootstrap, 1000 replicates. Block sizes 2/10/20 cells $\approx$ 1/5/10 km.

Region	Block ($\approx$ scale)	Baseline AUC	Thermal AUC	Δ AUC	Δ AUC 95% CI
Manavgat 2021	2 ($\sim$ 1 km)	0.803	0.870	+0.067	[+0.055, +0.079]
	10 ($\sim$ 5 km)	0.748	0.797	+0.050	[+0.023, +0.077]
	20 ($\sim$ 10 km)	0.683	0.731	+0.048	[+0.014, +0.085]
Bejís 2022	2 ($\sim$ 1 km)	0.862	0.918	+0.056	[+0.048, +0.065]
	10 ($\sim$ 5 km)	0.779	0.825	+0.045	[+0.018, +0.069]
	20 ($\sim$ 10 km)	0.739	0.795	+0.057	[+0.031, +0.090]
Muğla 2021	2 ($\sim$ 1 km)	0.743	0.859	+0.116	[+0.106, +0.125]
	10 ($\sim$ 5 km)	0.698	0.777	+0.079	[+0.050, +0.105]
	20 ($\sim$ 10 km)	0.673	0.733	+0.061	[+0.030, +0.094]
North Evia 2021 (ext.)	2 ($\sim$ 1 km)	0.759	0.912	+0.153	[+0.142, +0.166]
	10 ($\sim$ 5 km)	0.716	0.864	+0.148	[+0.119, +0.183]
	20 ($\sim$ 10 km)	0.679	0.833	+0.154	[+0.124, +0.189]
Montiferu 2021	2 ($\sim$ 1 km)	0.781	0.883	+0.101	[+0.080, +0.125]
	10 ($\sim$ 5 km)	0.620	0.720	+0.100	[+0.017, +0.186]
	20 ($\sim$ 10 km)	0.555	0.681	+0.126	[+0.053, +0.228]

490 **20-cell row of this table should be read as indicative rather than as**
an interval. The 10-cell row is the coarsest blocking this design supports properly, every region there having 16 to 70 positive-carrying blocks, and the increment holds at that scale in all five regions. Source: `paper/referee2_numbers.md`, block C.

495 This within-region result is not itself novel. It is reported because the transfer arms below are measured against it.

4.3. What the evaluation frame is worth, within one region

This section isolates the effect Section 4.4 applies between regions. It comes first because it is measured on one model in one region with no transfer
500 involved, so nothing the transfer arms do can explain it.

Where the skill is lost, on a matched comparison. Four evaluations are reported. The last three are scored on **identical cells**, so they differ only in what the model was trained on; the first shows why an unmatched comparison misleads. Each uses the transfer protocol unchanged: fit, then
505 apply with no refit, no recalibration and no threshold selection. The held-out unit is a burned connected component of at least 50 cells together with all cells within 2 km of it.

What makes that a harder problem than a whole region is the composition of its negatives, not its burned fraction, and that can be shown rather than
510 argued. Taking the same fitted model and the same out-of-fold predictions and scoring them on a random sample of region cells drawn at the scar area's own burned fraction gives **0.782 against the region-wide 0.782**: matching the prevalence changes nothing, at -0.000 $[-0.003, +0.002]$ over the nine scars. Scoring the same predictions on the scar area itself gives

515 0.627, a fall of **+0.155** [**+0.093**, **+0.217**]. The whole effect is the negative pool. Every negative in a scar collar is fire-adjacent, sharing the terrain, land cover and synoptic conditions of the positives, whereas a region’s negatives include its easy far field. The burned fraction, 34 to 87 % against 3.8 to 28.7 % for a region, is a symptom of that construction, and ROC-AUC is in any
520 case invariant to class balance at fixed class-conditional distributions.

Table 2: The four evaluations, scored on identical cells. Primary natural-vegetation population. Rows B, C and D are scored on the held-out scar area; row A is the whole region and is shown to make the mismatch visible. Means and Student *t* intervals are over the eight held-out scars.

Evaluation	Model trained on	Scored on	Mean AUC	95 % CI
A. Blocked cross-validation, 5 km	the region, scar included	the whole region	0.776	[0.738, 0.814]
B. Same blocked model, restricted	the region, scar included	the scar area	0.634	[0.552, 0.716]
C. Leave-one- scar-out	the region, scar withheld	the scar area	0.552	[0.501, 0.602]
D. Foreign region	another region, 306 to 2,802 km	the scar area	0.555	[0.495, 0.616]

All four rows are means over the **same eight scars**. A ninth burned component, Bejís, is excluded throughout: it is that region’s only component of any size, so holding it out leaves no usable source model and there is no row C for it. Rows A, B and D are reported here on the eight so that the
525 differences are paired, and the intervals are Student *t* over those eight, which is the resampling unit for this arm rather than the spatial-block bootstrap used elsewhere in the paper. Eight is a small number and the intervals are

wide accordingly. Four of the eight scars are in Muğla and two in Montiferru, so they are not independent; row A in particular is a region-level quantity
530 repeated across the scars of a region, and its interval is pseudo-replicated and should not be read as coverage. The two differences are computed per scar and paired, which is what they are reported for.

The same model, scored two ways on the same region, differs by 0.143 AUC. Rows A and B are one model. The only change is which cells
535 it is scored on: the whole region, or the burn scar and its 2 km collar. That change alone costs **0.143** [**+0.077**, **+0.208**] of the 0.225 [**+0.157**, **+0.293**] fall from A to C, about two thirds, and it is the size of the entire increment this literature usually reports. Both intervals are Student t over the eight scars, so the "two thirds" is a ratio of two estimates and is compatible with
540 anything from roughly a third to nine tenths; the point of the comparison is the size of the numerator, not the precision of the fraction. A region-wide blocked figure is therefore an upper bound on what the same model achieves where the fire actually is, and the difference is not small enough to ignore. Any comparison of a scar-level result against a region-level reference inherits
545 that, and the region-level reference is what a paper of this kind normally reports.

Neither withholding the fire nor moving 2,800 km has a measurable cost. B minus C, the fire-specific residual on identical cells, is **+0.082** [**-0.011**, **+0.175**], which spans zero. C minus D, the effect of replacing
550 a same-region model with one fitted between 306 and 2,802 km away, is **-0.003** [**-0.075**, **+0.069**], which also spans zero and is an order of magnitude smaller. The second is the cleaner result: on the same cells, a model

fitted in another country does as well as one fitted in the same region with the scar withheld, and it survives restriction to Muğla, the only region where
555 holding out one scar still leaves the source model properly trained, at 0.579 against 0.597.

Row D is a mean over four foreign models, and the spread behind it is large. Decomposing it gives **32** scar-by-source combinations running from 0.374 to 0.724, with **ten below chance** and a mean spread of
560 0.171 across the four sources for a given scar (Appendix A, Table A3). The pooled 0.555 is therefore not a statement that any foreign model does as well as a same-region one, but that the *average* one does, and the variation it averages over is exactly the pair-specific instability that the rest of Section 4 is about.

565 One confound must be stated. Holding out a region's only large scar also removes most of its positives: the source model retains 11 positives in Evia, 88 in Manavgat and 97 in Montiferru, against about 2,000 in Muğla, which has four separate scars. Those four non-Muğla arms average 0.525 and the four Muğla arms 0.579. Three of the four are genuinely starved
570 — the fourth, Montiferru's second component, retains 472 source positives — so row C mixes "the fire was withheld" with "almost all the positives were withheld". Restricted to Muğla, C is 0.579 and D is 0.597, and the C-to-D comparison still shows nothing. **With eight scars, three of them starved, this design cannot establish a fire-specific residual, only**
575 **bound it at about 0.18.**

What this establishes is bounded rather than positive. Rows C and D both sit close to chance, so the comparison between them has little dynamic

range, and neither the fire’s identity nor the region boundary is shown to cost anything here. What is measurable is the change of evaluation geometry
580 between rows A and B. Nor does it establish the fire event as the unit, because the held-out patch is defined by the labels, so its identity cannot be separated from its location. The within-region half-split, which does see half of the target scar, returns 0.574 against leave-one-scar-out’s 0.552, with overlapping intervals.

585 **The result is robust to how the held-out patch is defined**, at 0.543 to 0.565 across minimum component sizes from 25 to 200 cells and under both 4- and 8-connectivity, and flat across buffers of 2, 5 and 10 km.

The increment declines with the holdout, and is not established once the fire is withheld. Contribution 1 is about the paired thermal-
590 minus-baseline difference, so the same evaluations were run on it.

Evaluation	Thermal increment	Interval
Blocked cross-validation, per region	+0.056 to +0.153	every interval above zero
Within-region half-split	+0.027	positive in 13 of 18 splits
Leave-one-scar-out	+0.022	[−0.032, +0.077]
Cross-region, twenty directions	+0.004	[−0.028, +0.036]

The point estimate falls monotonically as the holdout hardens, and the last two intervals span zero. Under leave-one-scar-out the increment is positive in six of eight scars and averages +0.022, against a baseline of 0.529 and
595 a thermal 0.552 that are both close to chance on that evaluation; restricted to Muğla it is +0.009 [−0.051, +0.070]. **The within-region increment**

of **+0.056 to +0.153** is therefore substantially a property of interleaved holdout. It is not shown to be zero on an unseen fire, and eight scars cannot show that; what this design establishes is that it is not established there.

4.4. The same effect between regions, applied to our own matrix

Section 4.3's effect applies with equal force between regions, and this section reports that test before the transfer matrix rather than after it, because it changes what Sections 4.5 and 4.7 can claim. It withdraws nothing in Section 4.3, which is what the test is built from. Sources: `aoi_frame_auc.csv`, `aoi_frame_transfer.csv`, `collar_frame_bootstrap.csv`, `diagnostics_collar_frame.csv` and the code under `paper/code/`.

The five areas of interest are not comparable frames. Each is a rectangle drawn around a fire, and they differ by an order of magnitude in how much unburnt far field they enclose: the share of modelled cells beyond 10 km of any burned cell runs from **2.1 %** in Montiferru to **63.1 %** in Bejís (Appendix B, Table B6). That far field is not a neutral addition — in Manavgat the median elevation of modelled cells rises from 472 m within 5 km of the fire to 1,273 m at 20 to 50 km, against 512 m for the burned cells themselves.

Under an equalised frame the sign reversals of Section 4.7 do not survive. Restricting every region to cells within 10 km of any burned cell removes only far-field negatives; every burned cell is at distance zero and is retained at any radius, so the protection against choosing a flattering radius is the sweep below, not the retention of positives. On the equalised frame all five regions agree in sign on elevation, on LST and on TVDI, and

both bootstrap-supported elevation reversals of Table B3 disappear. Under this paper’s own criterion from Section Appendix C.7 — each region’s own 10-cell block-bootstrap interval must exclude 0.5 — **no reversal remains**
625 **bootstrap-supported.** Two features straddle 0.5 at the point estimate and neither is supported: `1st_anomaly_mean` at 0.392 [0.324, 0.460] in Bejís against 0.584 [**0.497**, 0.669] in Evia, where Evia’s interval includes 0.5 by 0.003; and `tvdi_difference_mean` at 0.509 [0.417, 0.597] in Muğla against 0.384 [0.288, 0.504] in Montiferru, where no region’s interval excludes 0.5.
630 These are the two internally differenced channels. An earlier draft treated the first as uniquely informative because it carries no lapse-rate signal; that does not hold, since the second is equally decorrelated from elevation and behaves the same way.

A weaker instrument does support the anomaly result. Table B3’s
635 note commits this paper to a difference interval on the pair as the sharper test. Applied under the collar, four pairs have opposite-sided point estimates **and** a difference interval excluding zero, all on `1st_anomaly_mean` and three of the four involving Evia (Appendix A(1), Table A6). Three caveats keep it weak: nine of ninety feature-by-pair differences clear zero against about 4.5
640 expected under the null with no multiplicity correction, the nine features are effectively two to three dimensions, and Evia’s own interval-support status turns on 0.003. **The honest statement is that no reversal meets this paper’s strict criterion under the collar, and that the LST anomaly differs between regions on the weaker difference instrument.** Eleva-
645 tion under the collar is above 0.5 in all five regions but individually supported in only two.

The sign the five regions agree on is not the one the dryness
 framing predicts. In every region a hotter pre-fire surface is associated
 with less burning, and the same holds for TVDI. It is neither a lapse-rate
 artefact nor greenness acting through fuel load: on mutual adjustment the
 surviving channel is LST, not NDVI. Two caveats bound it — in two re-
 gions it is a residual spatial gradient that disappears when distance to the
 nearest burned cell is stratified inside the collar, and interval support is not
 uniform, so "all five agree" is a statement about point estimates. What
 we can defend is that the absolute thermal channels behave here as **static**
land-surface descriptors rather than as a dryness index, and that the two
 internally differenced channels carry no consistent cross-region direction at
 all. Compositing depth was not tested and remains an open alternative. Per-
 region values are in Appendix A(k); Section 5.2 states the consequence for
 the moisture-stress motivation of Section 1.2.

The same test determines what the diagnostics of Section 4.6
 can establish. The only two candidates there with intervals excluding zero
 measure agreement in the sign of each predictor's association between source
 and target, counted over features interval-supported in both. Those are built
 out of exactly the signed AUCs this section has shown to be frame artefacts,
 and in Section 4.6 they are correlated against transfer measured on the same
 unequal frames. Recomputing both sides under the 10 km collar:

The two fail differently and both fail. Once frames are equalised every
 region pair agrees in sign on every jointly supported feature, so the agreement
 fraction has no variance left. The supported cosine keeps a trace of variance
 — unanimous signs fix directions but not magnitudes — and simply stops

Table 3: The diagnostics that ordered transfer, recomputed on an equalised frame. Spearman ρ against target ROC-AUC over the ordered directions in which each is defined. The first two rows are the two variants that cleared zero in Table B1; the third is the all-feature cosine, which did not and is shown for contrast. Both sides are recomputed here under one bootstrap setting, 1000 replicates, seed 42. Source `diagnostics_collar_frame.csv` and `collar_increment_and_cosine.csv`.

Diagnostic	Full frame	10 km collar
Sign-agreement fraction, supported features	$\rho = +0.86$ ($p = 0.0001$, $n = 14$)	1.0 in all 18 directions, variance exactly 0 — degenerate
Cosine, supported features	$\rho = +0.81$ ($p = 0.0005$, $n = 14$)	$\rho = -\mathbf{0.06}$ ($\mathbf{p = 0.82}$, $\mathbf{n = 18}$), variance 0.00014
Cosine, all nine features	$\rho = +0.50$ ($p = 0.023$, $n = 20$)	$\rho = +0.12$ ($p = 0.61$, $n = 20$)

tracking transfer. This is not a marginal shift: features supported in both regions *rise* from 1.20 to 3.40 per direction, so the diagnostics are better determined and unanimous. The disagreements they were reading were the
675 far fields.

Two scope statements belong with the table and are given in Appendix A(p): the full-frame values here are not numerically the published ones, because the support test is itself bootstrap-dependent; and only these diagnostics were recomputed on the collar, so the other eighteen are unknown
680 against the equalised transfer vector rather than shown to be null.

This settles what Contribution 3 can claim. It is not that conditional similarity orders transfer where marginal similarity fails; it is that **no diagnostic tested here has been shown to order transfer**. The eighteen marginal, niche and regime candidates do not order it on the frames as drawn,
685 and the two conditional variants that do stop doing so once the frames are comparable, because what they were reading is how the study rectangles were drawn. The practitioner’s position is therefore worse than the as-drawn analysis suggests, not better: there is no screen, and the apparent exception is an artefact. Section 4.6 reports its numbers as computed under the
690 pre-registered protocol, which is what a reader following that protocol would obtain.

The same-geography arm of Section 4.10 does not survive either, and it is the most extreme case in the cohort. A fixed study area is not a fixed evaluation frame: its 2022 arm has 93.2 % of cells beyond 10
695 km of any burned cell against 55.3 % for 2021. Under the same collar the 2021 figure barely moves while the 2022 figure crosses to the same side of

0.5 with an interval covering chance, so **no bootstrap-supported sign reversal survives anywhere in this paper once evaluation frames are equalised**, between regions or between two fires in one region. Values
700 in Appendix A(m).

The within-region increment survives the same correction, remaining positive in all five regions on the equalised frame with a mean of +0.077 against +0.086 as drawn, and halving but staying positive in all five at a 5 km collar. The paper’s one surviving predictor-level positive claim
705 therefore holds on the frame it argues is the correct one. Per-region values are in Appendix A(r).

The transfer matrix moves as well. Restricting source and target to the same collar:

The reference arm reproduces the frozen matrix, at 0.540 against Table
710 B9’s 0.541 and 14 of 20 exactly, so this is measuring the same quantity. **The baseline control moves with it and must be restated on this frame:** the static baseline transfers at 0.593 against the thermal model’s 0.617, a paired difference of +0.023 rather than the +0.003 of the frame as drawn. The control still holds in kind — the static predictor class is not the portable
715 one either — but the gap between them is eight times larger once frames are comparable, and Sections 1, 5 and 6 quote only the as-drawn pair. The above- and below-chance counts in this table are point counts. Under the same 10-cell block bootstrap used for Table B9, at 1000 replicates, the full frame gives nine directions above chance and four below with interval support, and the
720 collar frame fifteen above and one below, so the headline movement is six to one at the point estimate and **four to one with interval support** (aoi_

Table 4: Cross-region transfer under equalised evaluation frames. Primary natural-vegetation population, thermal model, twenty ordered directions per row. Above/below chance are point counts; the supported counts use the same 10-cell (≈ 5 km) spatial-block bootstrap on the target as Table B9, 1000 replicates, seed 42. Per-direction bounds are in `aoi_frame_transfer.csv`.

Source frame	Target frame	Mean target AUC	Above chance	Below chance	Supported above / be- low	Pairwise thermal delta
full	full (Table B9)	0.540	14 of 20	6	9 / 4	+0.003
full	10 km	0.575	17 of 20	3	11 / 1	+0.002
10 km	full	0.571	17 of 20	3	11 / 3	+0.014
10 km	10 km	0.617	19 of 20	1	15 / 1	+0.023
5 km	5 km	0.608	18 of 20	2	12 / 0	+0.014

`frame_transfer.csv`, which carries the per-direction bounds). The largest movers are Bejís to Evia, 0.383 to 0.602, and Bejís to Manavgat, 0.440 to 0.601.

725 **What this settles, and what it leaves standing.** Five quantities reported in the sections that follow are properties of the frames rather than of the predictor-burning relationship, and are identified as such where they appear: the count of six anti-predictive directions, which becomes one; the sign reversal of elevation, LST and TVDI as a mechanism; the sign-agreement
730 diagnostic of Section 4.6; the same-geography arm of Section 4.10; and the paired thermal contribution, which is +0.004 as drawn and +0.023 equalised. What the correction leaves standing is the central negative result, and its size must be stated on a matched comparison. Setting the equalised transfer mean of 0.617 against a within-region reference of about 0.87 would compare a 10
735 km-collar number with a full-rectangle one at 1 km blocking, which is the most generous reference in the paper and the very figure Section 4.3 argues is an upper bound. Recomputing the within-region reference on the same frame and at the 5 km blocking this design defends (`matched_frame_gap.csv`, `paper/code/verify_matched_gap.py`):

Frame	Within-region (5 km blocking)	Mean transfer	Gap
full rectangle	0.798	0.540	0.258
10 km collar	0.772	0.617	0.155
5 km collar	0.737	0.608	0.129

740 Paired by target region, the collar shortfall is **+0.155** [**+0.094**, **+0.217**] (Student t over the five target regions; per-region +0.086 Montiferru, +0.127 Manavgat, +0.161 Muğla, +0.196 Bejís, +0.206 Evia), which reproduces the

point estimate above and excludes zero. The shortfall therefore survives
745 on every matched row, but it is **0.155 at the collar, not the 0.25 the
unmatched comparison implies**, and it shrinks as the frame approaches
the fire — which is where a susceptibility surface is actually used.

This is Section 4.3’s effect acting between regions rather than within one,
on frames whose fire-adjacent share ranges from 37 % to 98 %. Section 5.9(x)
750 records the frame as a limitation of this cohort rather than of the method.

4.5. Cross-region transfer, and what label-free adaptation does to it

**Everything in this section is computed on the frames as drawn,
and should be read against Section 4.4.** That section has already shown
the frames are not comparable, that equalising them lifts mean transfer from
755 0.540 to 0.617 and leaves **one** direction below chance rather than six, and
that the within-region reference used below is itself frame-dependent. The as-
drawn matrix is reported in full because it is what the pre-registered protocol
yields and what a reader following that protocol would obtain; the corrections
are not repeated at each number.

760 **Raw transfer is heterogeneous and includes anti-predictive di-
rections.** Raw target AUC spans 0.326 to 0.686 (Fig. C.4). Twelve of 20
directions are above chance with CI support. Six are *below* chance with CI
support: both directions of Manavgat to Bejís and of Manavgat to Muğla,
plus Bejís to Evia and Evia to Bejís — of which only Manavgat to Bejís sur-
765 vives frame equalisation. Two intervals span 0.5. Even the best raw transfer,
Evia to Manavgat at 0.686, stays far below that target’s own within-region
thermal performance of 0.870. Across all directions the raw deficit against
the within-region reference is 0.184 to 0.592 AUC. That reference is a region-

level blocked estimate, and Section 4.3 shows it is not matched to a transfer
770 evaluation.

Those counts belong to the 2-cell blocking of Table B9. At the more
conservative 10-cell blocking the same points give 9 above, 4 below and 7 un-
certain, and no direction changes side of the chance line (Section 4.9). Four
of the six below-chance directions keep their support there. Bejís to Man-
775 avgat and Manavgat to Muğla lose it and carry no verdict. The qualitative
statement is unchanged. The counts should not be read as exact.

**In precision terms the transfer is worse than the ROC fig-
ures suggest.** A susceptibility surface is used as a ranked area budget,
so precision-recall is the operational quantity. Across the twenty directions
780 the thermal model’s PR-AUC averages **0.156 against a no-skill baseline
of 0.136. Six of the twenty fall below their own no-skill baseline
at the point estimate, five of them with intervals entirely below it;**
the exception is Bejís to Manavgat, whose interval covers its baseline. Only
one direction exceeds twice its baseline. These are frame-as-drawn quantities
785 and the PR arm was not recomputed on the collar. Per-direction values are
in Appendix B, Table B4.

The static baseline does not transfer either. The same twenty
directions were run with the terrain, fuel and greenness baseline alone, giving
a mean target AUC of **0.537** against **0.541** for the thermal model. Static
790 attributes of a place are the class Dimarco et al. transfer successfully and the
class this paper’s framing treats as portable; here that class is itself barely
above chance. The transfer failure below is therefore not specific to the
dynamic block: a baseline that does not travel, plus pre-fire thermal state,

gives a model that does not travel.

795 **The thermal block’s paired contribution to transfer, with its interval.** Differencing the two matrices direction by direction gives a mean of **+0.004**, but the mean is not the informative statistic. The individual contributions span **−0.148 to +0.133**, twelve positive and eight negative, so the spread is thirty times the mean and the sign belongs to the pair rather than
800 to the block; a mean near zero records cancellation, not consistent absence of effect. The directions are not independent, since each region appears in eight of the twenty, so the interval depends on the resampling unit — **all four units the design permits give the same answer**, from $[-0.027, +0.034]$ treating directions as independent to $[-0.037, +0.046]$ jackknifing regions,
805 and none propagates within-direction sampling variability (Appendix A(q)). The leave-one-region-out jackknife shows how little the mean is anchored: **dropping Evia alone reverses its sign.**

On the frames as drawn, **six directions are below chance with interval support**, the sharpest at 0.326 $[0.305, 0.349]$. Section 4.4 has
810 shown most of that count is a property of the frames: equalising them leaves **one**, Manavgat to Bejís at 0.417 $[0.349, 0.488]$, supported at the 10 km collar though its interval covers chance at 5 km. That direction still needs a mechanism acting on the direction of the relationship, because no account of merely lost skill produces a reliably reversed ranking, and Sections 4.6 to
815 4.10 pursue it. Everything below this point is computed on the frames as drawn and should be read against Section 4.4. Per-split and per-scar detail is in Appendix A(i); per-direction values for all twenty directions, raw and under both adaptations, are in Appendix B, Table B9.

Label-blind adaptation compresses the matrix toward chance
 820 **rather than repairing it.** Under region-wise z-scoring the twenty direc-
 tions span 0.431 to 0.630 and under CORAL 0.443 to 0.624, roughly half
 the raw spread, with no adapted direction exceeding 0.631 against the un-
 matched within-region references of 0.859 to 0.918. Adaptation raises the
 failing directions and degrades most of those that already transferred. Tak-
 825 ing the better of the two adaptations per direction, 14 of the 20 end closer
 to chance than they began and 6 end further from it; five of those six involve
 Montiferru, the smallest and last-added region, and move upward, while the
 sixth is Manavgat→Muğla moving downward from 0.470 to 0.443. The 14 to
 6 split should be read at the precision of limitation (ix) in Section 5.9, since
 830 Bejís→Manavgat is counted as compressed on a margin of 0.001.

The per-direction decomposition is in Appendix A(j), Table A5; two fig-
 ures from it matter here and are used below.

Against the right reference, adaptation is not failing (Fig. C.5).
 The three controls above give an achievable reference for a model applied to
 835 a fire it has not seen: 0.574 for the half-split, 0.552 for the leave-one-scar-out.
 CORAL, committed to in advance, averages **0.552** across the twenty direc-
 tions against 0.541 raw, which is exactly that reference. Taking whichever
 of the two adaptations scores higher per direction gives 0.556, but that se-
 lection uses the target labels the protocol forbids, so it is an oracle upper
 840 bound rather than an achievable result and is reported as one. What it does
 is regress the matrix towards that value: fourteen of twenty directions move
 closer to chance, which harms the directions that already worked and helps
 the ones that did not. The verdict that follows is narrower than "alignment

fails". Alignment cannot exceed what a model can achieve on an unseen fire,
845 and it does not.

In the six directions where raw transfer was below chance, the best label-free method recovers at most 34 % of the gap to the within-region reference, so the remaining unrecovered fraction is at least 0.66 everywhere. The half-split control above shows that the larger part of that remainder is already
850 incurred inside the region, so it should not be read as a measure of concept shift. Seven directions show *negative* recovery, meaning adaptation moves the score away from the reference; in six of those raw transfer was already above chance and adaptation destroyed that advantage. Label-free alignment therefore does not act as a repair mechanism.

855 4.6. *Transferability diagnostics: what appears to order transfer, and why it does not*

Twenty candidate diagnostics from five families were each rank-correlated with the same target quantity, the raw thermal transfer AUC over the twenty ordered directions, under one common pair-based bootstrap.

860 Geographic separation does not order the matrix on either construction: over all twenty directions the Spearman correlation between centroid separation and transfer is -0.32 , and on the twelve-direction common subset -0.24 , both spanning zero. The two nearest directions are among the worst on the frames as drawn, while the 2,802 km pair returns 0.326 and 0.444, so
865 the transfer mean of 0.541 is not the value at any one separation.

Only two diagnostics have intervals excluding zero, and both are conditional: the sign-agreement fraction over interval-supported features at $\rho = +0.84$ $[+0.58, +0.88]$, and its cosine variant at $+0.81$. No marginal

Table 5: Transferability diagnostics versus raw thermal transfer, by family. Spearman ρ against raw transfer AUC with pair-based bootstrap 95 % CIs. Exp. is the sign expected if the diagnostic orders transfer. The member named is the one with the largest absolute correlation in its family, which is not always in the expected direction. All twenty individual diagnostics are in Appendix B, Table B1.

Family	Diagnostics	Exp.	Largest correla- tion in family	Sign as expected	Any CI excluding 0
P(y x) condi- tional	6	+	+0.84 [+0.58, +0.88] agree- ment fraction, supported features	yes	yes, 2 of 6
P(x y=1) niche overlap	5	+	+0.24 [−0.45, +0.74] Schoener’s D, 1-D mean	yes	no
P(x) marginal	6	−	−0.32 [−0.78, +0.33] domain- classifier AUC	yes	no
P(y) regime structure	2	−	+0.29 [−0.38, +0.74] ^{log} ₄₂ effective- N distance	no	no
geographic	1	−	−0.24 [−0.84,	yes	no

measure was shown to order the matrix, including area-of-applicability dis-
870 similarity, climatic distance and geographic distance, and neither were the
niche-overlap and regime families. The learned domain classifier is at ceil-
ing, separating source from target at $AUC \geq 0.96$ for every pair; it always
succeeds, which is why it carries no ordering information.

Four limits are stated with the result rather than after it. **Size:** the
875 index’s tie structure caps the achievable Spearman at +0.861, so the observed
+0.840 sits on that ceiling, and its exact one-sided permutation p of 0.0060
is the smallest that structure can produce against a Bonferroni threshold of
0.0026 over nineteen computed variants — **no outcome of this diagnostic
could have cleared family-wise correction on ten effective pairs.**
880 **Labels:** signed associations need burned labels in both regions, so the family
that appears to work is not available before deployment while the family that
fails is. **Selection,** the sharpest of the three: the two rows that clear zero are
the *supported-feature* variants, whose subset is chosen by whether two regions’
bootstrap intervals happen to be disjoint — a data-dependent selection made
885 on the same data, with no correction. Their unselected counterparts over all
nine features are $\rho = +0.50 [-0.17, +0.83]$ and $+0.18 [-0.40, +0.72]$, both
spanning zero, so the result lives in the selection step and is reported as such.

The families sit on unequal samples — twelve directions for the marginal,
applicability, climatic and geographic rows, sixteen for the supported-conditional
890 rows, twenty for the rest — so every row was recomputed on the common
twelve. Published values reproduce to 4.8×10^{-5} , the conditional rows still
lead at $+0.87 [+0.65, +0.88]$ and $+0.85 [+0.43, +0.88]$, and every marginal
row still spans zero. The ordering is not an artefact of unequal samples.

A fourth limit, established in Section 4.4, removes the result
895 **entirely**. The index is built from signed associations that Section 4.4 shows
to be frame artefacts, and was correlated against transfer measured on the
same frames. Recomputed on an equalised frame it is unanimous, with no
variance left to correlate. Everything in this section is what the original
protocol yields; the conclusion that survives is that **no diagnostic tested**
900 **here was shown to order transfer once the frames are comparable**.

4.7. The contrast pair: similarity is not sufficient

The clearest single view needs no ranking at all (Fig. C.8). Manavgat
and Muğla lie in the same country and fire year, 306 km apart by centroid,
and their burned cells occupy the most similar environmental envelope of any
905 pair in the matrix; Bejís and Montiferru occupy the least similar.

Per-quantity values for both pairs, on both frames, are in Appendix B,
Table B10.

Two readings of this pair do not survive Section 4.4 and are not offered.
On the frames as drawn five of nine feature-response directions point opposite
910 ways, elevation among them; under the collar that figure moves to 0.561
against 0.606, on the same side of 0.5. And transfer is below chance in both
directions as drawn but above chance under the collar. **What survives**
is the ordinal contrast: the most similar pair is among the weakest in
the matrix and the least similar among the stronger, while neither is the
915 extreme — the weakest direction is Manavgat to Bejís at 0.417 and the
strongest Muğla to Evia at 0.727. The claim this supports is that high
envelope overlap does not buy transfer, not that it produces anti-prediction.
The AoA shares are full-frame quantities.

Bejís and Montiferru sit at the opposite extreme. Their burned envelopes
 920 barely overlap, and they carry the most dissimilar values on every overlap
 measure. They transfer above chance in both directions. That half of the
 contrast rests on point estimates: neither direction carries a verdict at 5
 km blocking, and the marginal applicability audit was never produced for
 Montiferru. The claim is one of *sufficiency*. Similarity does not guarantee
 925 transfer, and dissimilarity does not preclude it. The claim rests on two
 coexisting counterexamples, so it does not depend on the number of pairs
 available.

4.8. Interventions: pooling and feature removal

Two interventions were pre-registered. **Pooling** four regions into one
 930 training set does not recover what single-source transfer loses: every pooled
 target still falls short of its own within-region reference, and only Bejís is be-
 low chance with interval support. **Removing the two predictors whose
 signed association reverses** costs -0.081 of mean within-region AUC,
 supported in every region and roughly three quarters of it attributable to
 935 elevation, a *baseline* terrain variable, and returns $+0.014$ $[-0.017, +0.045]$
 on transfer, whose interval spans zero. A local cost is measured; no com-
 pensating transfer gain is, so no exchange between the two is demonstrated.
 Both predictors were chosen because they reverse, using the same data on
 which the two figures are then estimated, and no correction for that selection
 940 is applied; Section 4.4 additionally withdraws the support for the selection
 rule itself. Per-configuration values are in Appendix A(n).

4.9. Sensitivity analyses

Eight design choices were varied with everything else held fixed. They are the Evia AOI and its prevalence, the CORAL regularisation constant, the blocking scale, the closure date of the predictor window, the quality
945 screening of the coarse thermal input, the contrast between the normalised and the absolute dryness channels, the removal of the coordinate-informed channels, and the capacity of the classifier. None changes a conclusion above. Two bound how the results should be read, so they are carried into the main
950 text here.

Coarsening the blocks from 1 km to 5 km moves the **paired thermal-minus-baseline delta** verdicts from ten positive, seven negative and three uncertain to six, four and ten. The above-chance verdicts on the thermal arm itself, a different quantity, move from twelve, six and two to nine, four
955 and seven. Support is removed from seven verdicts and added to none. The point estimates are unchanged, but that is an identity rather than a result. The blocking scale is the resampling unit, and it cannot move an estimate computed once over all target cells.

Manavgat’s whole downstream chain was then rebuilt from a quality-
960 screened MODIS input. That changes the downscaled surface on 22,304 of 24,150 cells, by up to 10.9 °C. No signed univariate association moves by more than +0.0003. This closes the one processing-artefact candidate for that region’s behaviour. Appendix A reports all eight arms, including one that tests a claim of Section 1.2 and does not uphold it.

965 *4.10. The same geography, a second fire*

Muğla burned twice, eleven months apart, on the same grid and through the same processing chain. The arm was designed to hold place fixed and vary only the fire, which would have separated regional concept shift from everything else that differs between study areas. On the frame as drawn it shows a bootstrap-supported elevation reversal between the two fires; Section 4.4 withdraws that, because the 2022 arm carries the largest far field in the cohort and under a 10 km collar the two arms fall on the same side of 0.5. The arm, its structural asymmetries and the direction of the bias they impose are in Appendix A(m).

975 *4.11. Distance within a region, and why it does not reframe the result*

One further arm measures how skill decays with distance inside a single region. A model is fitted on one half of a region and applied to the other, and target cells are binned by their distance from the training cells (`distance_curve.md`, `paper/code/distance_curve.py`). Means are unweighted over bins, whose positive counts range from 1 to 2,564, and the bins carry no intervals.

Separation from training cells	Bins	Mean target AUC
0 to 5 km	18	0.692
5 to 10 km	16	0.519
10 to 20 km	11	0.499
20 to 40 km	6	0.445
40 to 80 km	3	0.541
80 to 160 km	1	0.421
cross-region, 306 to 2,802 km	20	0.541

By 10 to 20 km inside a single region the model is already at
chance. That bounds how far a susceptibility surface of this kind can be
985 carried from the cells it was fitted on, and it is consistent with Section 4.3,
where withholding a scar and replacing the model with a foreign one cost
nothing distinguishable.

**It does not reframe the paper’s negative result, and a reading
that it does was considered and rejected.** That reading argued that
990 because the twenty cross-region directions average 0.541, at or above the
within-region plateau, they sit on the continuation of the curve and the fail-
ure needs no regional mechanism — a rule fixed in advance in `positive_
control.md`. Two objections defeat it, and `scar_control.md` records the re-
framing as withdrawn in full. **The rule cannot fail:** once the curve reaches
995 the chance floor, any cross-region mean near 0.5 lies on its continuation by
construction, so the comparison could not have come out otherwise, and a
test that cannot fail is not evidence. And **extrapolating an uninforma-
tive model does not produce a reliably reversed ranking:** Manavgat
to Bejís is below chance with interval support on the frame as drawn and at
1000 the 10 km collar, which is not what a model that has merely run out of skill
returns.

Four limits bound even the descriptive reading. The two distance ranges
do not overlap — within-region separations span 2 to 86 km, cross-region
separations start at 306 km — so any comparison across the gap extrapolates
1005 the curve. The far bins are thin, six at 20 to 40 km and one beyond 80 km, so
their means should not be read closely and the apparent rise at 40 to 80 km
is not evidence of anything. The near bins carry exactly the autocorrelation

blocked validation exists to remove, so **0.692 is an upper bound on near-field skill rather than an estimate of it**. And this design separates
1010 distance from crossing a study-area boundary, but not from the land cover, terrain and fire history that covary with it.

The arm therefore contributes a length scale for the within-region decay, not an attribution. The unit that fails to transfer is not established by this design, and Section 4.3 says so directly.

1015 *4.12. What target labels cost: the recovery curve*

Everything above measures a failure; this prices it. Label-free alignment does not close the residual gap, so the missing resource is information about the target that alignment cannot synthesise, and the direct way to supply it is target labels. A frozen labelled-budget diagnostic answers how many,
1020 for three regions across all six ordered directions, using one 10-cell (~ 5 km) spatial block as the unit of labelling effort and reading recovery against a matched target-only ceiling of 0.777 to 0.824.

Thirty-two labelled blocks recover **85 to 89 % of that ceiling in three of the six directions**, two of which started below chance; 51 to 57 % in two
1025 more, the directions where the conditional gap is widest; and 30 % in the sixth. That budget is 7 to 20 % of the target's natural-vegetation population, which at the 2,700 to 3,000 labelled cells those blocks carry (S1.3) and this grid's effective cell area of 0.199 to 0.208 km² (Appendix C) is roughly **540 to 620 km²**, so it is a real answer and not a cheap one. Two properties
1030 of the measurement bound it and are stated in the supplement rather than buried: at the top budget the selection pool is nearly exhausted, so the narrow upper-budget intervals reflect saturation rather than precision, and

the labelled blocks are drawn from the event being predicted, which is not a resource available before that event burns. The protocol, the full curve and
1035 the limits are in Supplement S1.

Two limits belong with the number. At small budgets the same intervention damages the direction that already transfers best without any labels. And six directions in three regions cannot support a general label budget, so none is offered.

1040 5. Discussion

5.1. Reading the three findings together

Section 1.4 states the three findings and Section 4 establishes them; this section argues from them rather than restating them. One relation between them does need saying, because it is what makes the paper cohere: the first
1045 finding is not a caveat attached to the other two, it is the instrument that sets their size. Applied to our own matrix it withdrew three claims we had made, and what it left standing is a shortfall in transferred skill, not a reversed relationship.

5.2. Why the thermal increment is real but local

1050 The within-region increment and the transfer failure are measured at different separations, and Section 4.3 shows most of the difference between them is already present inside a single region. The increment is therefore local in a specific sense: it holds where held-out cells are interleaved with training cells, and most of it is gone once they are not, before the fire or the region
1055 changes. It is not an artefact to be explained away — it replicates in five

regions, survives coarsening to about 5 km with its interval intact, persists in the secondary population, and survives a predictor window closed up to two weeks earlier — but it is established under interleaved validation and not beyond it. The window-closure result is region-specific: the contribution
1060 strengthens in Bejís and Muğla, is flat in Montiferru and **weakens monotonically in Evia**, so what holds everywhere is survival, not improvement (Appendix A(d)).

The natural objection is that different Mediterranean regions are simply different systems, so a predictor meaning one thing in one place and another
1065 elsewhere is a comparison of two systems rather than instability. **We designed the two-Muğla-events arm to answer that objection and it does not answer it.** On the frames as drawn it looked decisive — region, grid and processing chain identical, static predictors identical cell by cell, elevation’s association reversing between the two fires — but Sections 4.4 and
1070 4.10 show it is the most extreme frame artefact in the cohort, and under the collar the two arms fall on the same side of 0.5.

The objection therefore stands unanswered, and three further confounds were never resolved in any case. Season and year are confounded, and **that confound cannot be resolved in this study area**: the events sit 42
1075 days apart in median burn day-of-year, neither year contains a second event at the other’s phase, and a calendar-matched arm would carry nine burned cells against a gate minimum of thirty. The population is not fixed either, the 2022 arm being the 2021 arm with the 2021 scar removed. And that arm rests on eleven positive-carrying 5 km blocks against this design’s own
1080 floor of sixteen. With one fire per region everywhere else, and the one arm

that held place fixed now withdrawn, **this cohort provides no evidence that the transfer shortfall is regional rather than event-specific**, and Section 5.9(v) is a substantive limit rather than a formality. Section 4.11 adds a length scale for the within-region decay — the model is already at chance by
1085 10 to 20 km from its training cells — but explains why that cannot be turned into an attribution: once the curve reaches the chance floor, any cross-region mean near 0.5 lies on its continuation by construction.

One conclusion survives from the other direction. On the frames as drawn the sharpest supported reversal belonged to elevation, a *baseline* terrain variable, and the static baseline transfers no better than the dynamic block.
1090 Whatever the shortfall is, it is not the thermal block’s peculiarity.

5.3. *Why the similarity-based diagnostics fail*

A diagnostic built on distance in predictor space asks one question: do the target’s predictor values look like the training data’s? That is not the
1095 question that matters when the failure is conditional. A target region can sit well inside the training envelope while the relationship between those predictors and burning points the other way. Manavgat to Muğla is exactly that case, with 0.875 of target cells inside the weighted area of applicability and transfer among the weakest in the matrix. On the frames as drawn that
1100 pair is below chance in both directions; on the equalised frame it is 0.551 and 0.510, above chance but still among the weakest in the matrix, while the least similar pair reaches 0.669 and 0.624 (Section 4.7). The point does not need anti-prediction: high overlap does not buy transfer.

The same holds for the niche-overlap and regime families. At the point
1105 estimates, the pair with the highest burned-niche overlap in the matrix

fails in both directions while the pair with the lowest transfers in both, so no monotone function of overlap can order the outcomes. The qualifier is not decorative: at the 5 km blocking this paper otherwise defends, neither Bejís↔Montiferru direction carries a verdict and Manavgat→Muğla loses its
1110 below-chance support, so this contrast is a statement about point estimates and is made as one. The domain classifier is at ceiling for every pair, which is informative in itself: the regions are trivially separable in predictor space whether or not their transfer works, so separability carries no ordering information here.

1115 *5.4. What the conditional diagnostic is, and what it is not*

The sign-agreement index was the paper's one diagnostic with an interval excluding zero, and Section 4.6 states three limits with the result: it sits essentially on its own tie-structure ceiling, it could not have cleared family-wise correction on ten effective pairs whatever it returned, and its feature
1120 subset is chosen on the same data, its all-nine-feature counterparts spanning zero. A fourth is that signed associations need burned labels in both regions, so it is a mechanism diagnosis rather than a pre-deployment screen.

A fifth limit removes it altogether. The index is built from signed associations that Section 4.4 shows to be artefacts of the evaluation frames, and
1125 recomputing it on an equalised frame leaves it unanimous and variance-free, while the second variant that cleared zero loses its correlation entirely, from $\rho = +0.81$ to -0.06 . So the practical conclusion is not "use this index instead", and it is not even "the index works but needs labels". It is that **none of the twenty candidates was shown to order transfer**, and the two that
1130 appeared to were reading how the study rectangles were drawn. That is a

worse position for a practitioner than Section 4.6 alone suggests, and it is the honest one. As always these are nulls on ten effective pairs: not shown to order transfer, rather than shown incapable of it.

5.5. *What the two interventions do and do not show*

1135 Both interventions show the same shape. Pooling four regions never beats the best single-source transfer for any target and stays 0.28 to 0.50 below the within-region ceiling, so aggregation does not manufacture the missing conditional information. Removing the reversing predictors costs -0.081 of within-region skill with interval support in every region, and changes mean
1140 transfer by $+0.014$ $[-0.017, +0.045]$, which is not distinguishable from zero.

That pair of numbers is easy to read as an exchange, and it is not one. The thermal block's own contribution to transfer is $+0.004$, with an interval spanning zero. Removing the reversing predictors returns $+0.014$, with an interval spanning zero. Both arms are null on the portability axis. What
1145 the interventions measure is a local cost and no compensating transfer gain. That is not a conservation law, and it is not a rate at which local skill can be sold for portability. No such rate is estimated here.

5.6. *The regime hypothesis, reported as it happened*

A regime-structure explanation was stated in advance and the data con-
1150 firmed the null. The regime-distance correlation has the wrong sign at the point estimate, the most regime-similar pair fails in both directions, and the most regime-different pair transfers above chance. One mundane explanation can be set aside: cutting Muğla, by far the largest population, to Manavgat's cell count and separately to its positive count leaves Muğla's transfer

1155 behaviour inside the subsampling range in both roles. The error was in the
hypothesised grouping, not in the data, and with ten pairs this cannot refute
regime typology [2] in general.

5.7. *The empirical contrast with Dimarco et al.*

Dimarco et al. [8] transfer successfully across a comparable Mediterranean
1160 design and we do not, and the two results are not in conflict. Their predictors
are attributes of a place, ours describe the state of a surface in one season,
which suggests that the relationship between domain similarity and transfer
success is predictor-class dependent.

That reading is one of at least two, and we cannot separate them here.
1165 Their response variable is human-driven **ignition**; ours is burned **area**. Ig-
nition likelihood is dominated by human access and activity, stable and
near-universal drivers, whereas burned extent is dominated by spread. A
predictor-class explanation and a response-variable explanation are there-
fore confounded in this comparison. Our own data speak against a simple
1170 predictor-class reading in any case: the static baseline transfers at a mean
of 0.537 here, so within this cohort the place-attribute class does not travel
either.

5.8. *Implications*

For practice, one number comes before any other. In precision-recall
1175 terms, which is how a susceptibility surface is used, transferred models aver-
age a PR-AUC of 0.156 against a no-skill baseline of 0.136, and six of twenty
directions fall below their own baseline on the frames as drawn. Whatever

the ROC figures suggest, a model moved to a region it was not fitted in does not usefully rank burned cells there.

1180 The paper supports one concrete change in reporting. Alongside a spatially blocked within-region figure, report skill on a held-out burn scar and its surroundings. On these five regions the two differ by about 0.14 ROC-AUC on the same model, the size of the effect such papers usually claim, so a blocked figure alone should be read as an upper bound. Transfer skill
1185 likewise has to be *measured* rather than inferred from similarity, since the pre-deployment diagnostics used for that inference did not order it here. Where a model must be moved, the resource that closes the gap is target labels: thirty-two labelled 5 km blocks recover 85 to 89 % of the target’s matched ceiling in three of six directions and 30 to 57 % in the rest. That is
1190 a real answer and not a cheap one, being 7 to 20 % of the target’s natural-vegetation population, and at small budgets the same intervention damages the direction that already transfers best.

For method development, the results bound what unsupervised alignment can be asked to do. Two label-free methods moved fourteen of twenty direc-
1195 tions towards chance; five of the six they moved upward involve Montiferru, the smallest region, and the sixth moves downward. The committed-in-advance CORAL arm reaches 0.552, and taking whichever method scores better per direction reaches 0.556 — but that selection uses the target labels the protocol forbids, so it is an oracle upper bound rather than an achievable
1200 result (Section 4.5). Even the oracle only reaches the reference a model can reach on an unseen scar (Section 4.3). Alignment is therefore not failing far below an achievable target; it is regressing the matrix onto that target, which

costs the directions that already worked. What it cannot do is exceed the reference, and a sign reversal is not a distribution mismatch that realigning
1205 inputs would repair.

5.9. Limitations

The transfer failure is a finding, not a limitation. The limitations are the boundaries on how far it generalises.

(i) **No meteorological covariates** enter the models, so we cannot say
1210 how local skill and portability would change with weather.

(ii) **The same-geography comparison covers one region only**, and there year, seasonal phase and population are confounded; **that confound cannot be resolved in this study area** (Section 5.2).

(iii) **All labels derive from a single burned-area product**, MCD64A1,
1215 and no second product covers 2021 and 2022 at this resolution.

(iv) **Evia remains the most imbalance-atypical population** even after the AOI extension.

(v) **Each region contributes one fire season**, so regional concept shift is confounded with event meteorology and, since the one arm that held place
1220 fixed is withdrawn, this design cannot attribute the shortfall to region rather than to event.

(vi) **Cross-region point estimates carry an implementation tolerance** of roughly ± 0.02 to 0.03 if the library version is not pinned.

(vii) **The diagnostic correlations rest on an effective sample of
1225 ten region pairs**, and both the successes and the failures of Section 4.7 should be read at that power.

(viii) **Manavgat’s atypical transfer behaviour remains unexplained;** three candidates were tested and none survives (Appendix C).

(ix) **The interval-support counts are less stable than the point**
1230 **estimates behind them:** the published split turns on a lower bound of -0.00045 , so the sign pattern is stable and the counts are not.

(x) **The five areas of interest are not comparable frames, and**
this cohort cannot fully repair it (Section 4.4). The collar radius is itself
a choice and 5 km and 10 km do not agree exactly (0.608 against 0.617),
1235 and the frames were fixed upstream in **repo/**, so we can restrict them but
not extend them. Any future cohort should fix the frame by an explicit
accessible-area rule [3] before any predictor is computed; we treat that as the
main design lesson of this paper.

(xi) **One classifier family.** The headline numbers use a random forest
1240 with unlimited depth, the configuration most able to encode local structure
and least able to extrapolate. Appendix A(h) shows the transfer result is not
an artefact of that choice: three further estimators, including a penalised
linear one, all land between 0.510 and 0.556 and all place fourteen of twenty
directions above chance. Those are point estimates without intervals, so the
1245 ordering among them is not claimed as a result. Other model families were
not tried, and a different inductive bias might behave differently, but within
this family the negative result is a property of the predictors rather than of
an unregularised estimator.

6. Conclusions

1250 Where a fire model is scored decides what it appears to know. Holding the
model, its predictors and its fitting fixed and changing only which cells are
scored, moving from a whole study region to the burn scar and its 2 km collar
costs **0.143 ROC-AUC**. A control isolates the cause as the composition of
the negative pool rather than class balance. The effect is the size of the
1255 predictor-block increments this literature publishes as findings, so a region-
wide validation figure should be read as an upper bound on what a model
achieves where fire actually occurs, and reported as one.

That correction is not only other people’s problem. Applied to our own
five-region matrix it withdraws five of our claims. The study areas enclose
1260 very unequal far fields, and equalising them to a 10 km collar lifts mean
transfer from 0.540 to 0.617, reduces the directions below chance, and leaves
no sign reversal supported under our own criterion — including the one arm
that held place fixed, two fires in the same study area eleven months apart,
which we had wrongly exempted because its geography was constant while its
1265 evaluation frame was not. What had looked like a reversed predictor-burning
relationship was mostly a statement about how five rectangles were drawn.

What survives is the central result. Pre-fire thermal dryness adds a real
and repeatable within-region increment in all five regions, surviving spatial
blocking at about 5 km and a predictor window closed up to two weeks before
1270 the first labelled burning. It does not travel: on matched frames and matched
blocking, equalised transfer of 0.617 falls 0.155 short of the within-region
reference, the paired cross-region contribution spans zero with a sign that
varies by pair, and the static baseline transfers no better than the dynamic

one on either frame, so this is not a peculiarity of thermal predictors. The
1275 shortfall also cannot be anticipated: none of twenty diagnostics from five
families was shown to order the matrix, and the two that appeared to do so
require target labels on both sides and stop ordering transfer once the frames
are comparable. Label-free alignment compresses most directions towards
chance rather than repairing them, and supervised recalibration works but
1280 is not cheap.

Two limits should be read with all of this. The agreed thermal direction
is the opposite of the one the dryness framing predicts — hotter pre-fire
surfaces burned less in every region, and holding greenness does not remove
it while holding temperature reverses greenness in two regions — so on this
1285 cohort the absolute channels behave as static land-surface descriptors rather
than as a dryness index. And with one fire per region, and the single arm
that held place fixed now withdrawn, this design measures a transfer shortfall
but cannot attribute it to region rather than to event.

Two practical consequences follow. Report local skill and portability sep-
1290 arately, since a predictor block worth a large within-region increment may
contribute nothing distinguishable from zero across regions. And fix the
evaluation frame by an explicit accessible-area rule before any predictor is
computed, because a frame chosen for convenience can manufacture both
a transfer failure and a mechanism to explain it. Three extensions are left
1295 for future work: temporal transfer, meteorological covariates, and physically
normalised dryness variables.

Appendix A. Appendix A. Sensitivity analyses

Each arm below varies one design choice and leaves everything else fixed. None changes a conclusion in the main text. They are reported so that a reader can see which choices were tested, and what each was worth. Arm (f) is the exception in one respect: it is not a robustness check but a test of a claim the introduction makes, and the claim is not upheld.

(a) Evia AOI and prevalence. The raw transfer arms were repeated with the legacy, high-prevalence Evia box. Every qualitative conclusion is unchanged. Thermal raw transfer AUCs move by up to 0.07, the largest being Evia to Bejís at 0.378 to 0.448, and no direction changes side of the chance line.

(b) CORAL regularisation, including the canonical $\lambda = 1$. Over nine λ values from 0 to 10^{-1} on four directions, CORAL transfer AUC moves by at most 0.014 within any direction and 0.008 within the thermal family. The canonical $\lambda = 1$ of the cited method lies outside that sweep and was computed separately on the same four directions. It changes the thermal mean from 0.519 to 0.522, a shift of +0.003 with a per-direction range of -0.005 to $+0.016$, and none of those four directions crosses the chance line. The $\lambda = 1$ values are 0.510, 0.444, 0.559 and 0.575 against 0.508, 0.445, 0.564 and 0.559 at $\lambda = 0.1$.

Those four directions are Bejís–Muğla and Manavgat–Muğla, and they exclude the pair that moves. An earlier version of this appendix generalised from them to the whole matrix, which was an error: the Manavgat–Bejís pair is not in the sweep and it is the pair whose CORAL result is λ -dependent. A separate frozen artefact, `experiments/cross_`

region/step10/coral_lambda_sensitivity.csv, carries it. Bejís to Man-
 avgat gives 0.5571 [0.5292, 0.5857] at $\lambda = 10^{-5}$, 0.5644 at $\lambda = 10^{-3}$ and
 0.5473 at $\lambda = 0.1$, all with intervals entirely above 0.5, but **0.4850 [0.4496,**
 1325 **0.5197] at $\lambda = 1$** , which crosses the chance line and loses its interval support.
 Manavgat to Bejís falls likewise, from 0.5108 to 0.4776 [0.4501, 0.5013].

The correct statement is therefore narrower. CORAL-dependent conclu-
 sions are insensitive to λ over three orders of magnitude below 0.1; at the
 cited method's canonical $\lambda = 1$ the one direction whose adapted interval sits
 1330 entirely above chance loses that status. $\lambda = 10^{-5}$ is the minimally regularised
 choice and matches the reference implementation, but the dependence is real
 and is reported rather than absorbed.

(c) Blocking scale. Recomputing the transfer quantities at 10-cell ($\approx$
 5 km) blocking from the frozen per-cell predictions widens the intervals and
 1335 moves the verdict counts, from ten positive, seven negative and three uncer-
 tain at 1 km to six, four and ten at 5 km. Coarser blocking therefore removes
 support from seven verdicts and adds none. Across five seeds the counts run
 5 to 6 positive, 3 to 4 negative and 10 to 11 uncertain, so they are not exact.

The point estimates are unchanged. That is an identity rather than a
 1340 result, and it should not be offered as robustness. The blocking scale is
 the bootstrap *resampling unit*, and each point estimate is computed once
 over all target cells, so no choice of block size could have moved one. The
 comparison does establish two things. Every verdict that changes, changes
 towards "no verdict". And no direction crosses the chance line under the
 1345 widened intervals. The counts are the fragile part of this paper. The sign
 pattern is not thereby shown to be robust. It is simply not tested by this

variation.

(d) Predictor-window closure. The predictor window was closed 7 and 14 days earlier in all five regions. The thermal contribution stays positive
1350 and bootstrap-supported everywhere. The direction of the change is region-specific. It strengthens in Bejís, from 0.058 to 0.079 at 14 days, and in Muğla, from 0.115 to 0.128. It is flat in Montiferru. It weakens monotonically in Evia, from 0.156 to 0.149 to 0.135. What holds in every region is survival, not improvement. That is the claim made in Section 5.2.

(e) Quality screening of the coarse thermal input. Two of the
1355 five regions' MODIS inputs are quality-screened and three are not. The split follows export date rather than design, and it induces an elevation-correlated change at the input, at $r = +0.615$ in Manavgat. Manavgat's entire downstream chain was rebuilt from a quality-screened input. That
1360 changes the downscaled surface on 22,304 of 24,150 cells, by up to 10.9 °C. No signed univariate association moves by more than +0.0003. Elevation is identical in both arms, at 0.374, the population is unchanged, and the within-region increment moves from $[+0.055, +0.079]$ to $[+0.054, +0.077]$. The reason is structural. Elevation is a DEM variable the screening cannot
1365 touch, and fusion falls back on the MODIS-derived surface across only 2.14 percentage points of coverage. Details are in `paper/modis_qc_downstream_propagation.md`.

(f) Normalised against absolute dryness channels. Section 1.2 argues that an internally normalised index should be less exposed to absolute-
1370 temperature offsets between regions than raw land surface temperature. The thermal block contains both kinds, so the argument can be tested directly as

a feature-set contrast.

One thing should be said before the result, because it makes the outcome less surprising than it might otherwise appear. The normalised channels are the two that hold their direction in the same-geography two-event comparison of Section 4.10, where the absolute channels move. That is a statement about one region across two fires. **Across regions it does not hold:** `1st_anomaly_mean` is itself one of the two predictors whose reversal is bootstrap-supported on the frames as drawn, reversing between Bejís and Evia (Section 4.4 withdraws that support under an equalised frame), which is why Section 4.8b drops it alongside elevation. Being internally normalised protects a channel against the offset between two seasons in one place. It does not, on this evidence, protect it against a change of place.

Three feature sets were run over all twenty directions and all five within-region folds, with the classifier, population, folds and bootstrap held fixed. The harness aborts unless its reference configuration lands on the frozen exports. It did exactly: the maximum absolute difference between the reference arm and the frozen transfer AUCs is 0.000000 across all twenty directions.

Feature set	Mean transfer AUC	Directions > 0.5	Mean within-region AUC
baseline only	0.5371	16	0.7896
baseline + normalised anomalies	0.5442	15	0.8551
baseline + absolute surface state	0.5479	13	0.8564
all ten features (reference)	0.5414	14	0.8883

1390 **The advantage is not found.** Per direction, the normalised set minus
the absolute set has a mean of -0.0037 . It is positive in 12 of 20 directions and
runs from -0.099 to $+0.094$, so the spread is an order of magnitude larger
than the difference. The normalised channels transfer very slightly worse
at the mean, and nothing here supports a design rule favouring anomaly-
1395 referenced dryness for portability.

Within region the two sub-blocks are indistinguishable, at 0.8551 and
0.8564, and each recovers most of the gap between the baseline’s 0.7896 and
the full model’s 0.8883. What separates them in sign stability does not
become transferable skill.

1400 This arm strengthens the negative finding rather than softening it. Four
differently constituted feature sets were tried. None clears a mean of 0.548
across regions, and all differ sharply within them. The baseline arm also
gives an independent confirmation of the control reported in Section 4.5: its

mean transfer of 0.5371 was recomputed here from the modelling datasets
1405 and matches the 0.537 read from the frozen per-direction export. Details are
in `paper/anomaly_only_transfer.md`.

The comparison is between two sub-blocks of one thermal set on one
cohort. It does not test normalised dryness indices in general, and it does
not test a normalisation fitted against a pooled multi-region reference rather
1410 than each region's own baseline years.

(h) Model capacity. Every number in this paper comes from one ran-
dom forest with unlimited depth and `min_samples_leaf = 3`. That is the
configuration most able to encode local structure and least able to extrapo-
late, so a reader may reasonably ask whether the transfer failure is a property
1415 of the predictors or of the estimator. Holding the model fixed is right for the
internal comparisons and does not license the external claim, so three fur-
ther estimators were run over the same twenty directions and the same five
within-region folds.

Estimator	Within-region AUC	Within increment	Transfer AUC	Transfer increment	Above chance
Random for- est, depth un- lim- ited, leaf 3	0.888	+0.099	0.541	+0.004	14 of 20
Random for- est, depth 6, leaf 50	0.829	+0.055	0.556	−0.007	14 of 20
Random for- est, leaf 200	0.805	+0.047	0.550	−0.021	14 of 20
Penalised lo- gis- tic re- gres- sion	0.741	+0.045	0.510	−0.024	14 of 20

1420

Regularisation does not rescue transfer. All four estimators land

between 0.510 and 0.556, all four put exactly fourteen of twenty directions above chance, and the linear model, the one built to extrapolate, transfers worst. Within-region skill falls as capacity is reduced, from 0.888 to 0.741, which is what regularisation is expected to cost; transfer does not rise to
1425 meet it. The thermal block’s contribution to transfer is +0.004 under the canonical forest and **negative** under all three regularised alternatives, so the block does not become portable when the model is made simpler.

The transfer failure is therefore a property of the predictors on this cohort, not of an unregularised forest. Detail in `paper/model_capacity.json`.

1430 (g) **The coordinate-informed channels.** `downscaled_1st_mean` and `fused_1st_mean` come from a per-region downscaling model whose own inputs include coordinates, which Section Appendix C.6 names as the one route by which a coordinate-derived surface re-enters a feature set that excludes coordinates. A coordinate-smoothed surface is by construction locally infor-
1435 mative and non-portable, so if the within-region increment depended on it, that increment would be this paper’s own result in miniature rather than a finding about thermal dryness. The arm was therefore run here rather than deferred.

Both channels were dropped and everything refitted, over all twenty direc-
1440 tions and all five within-region folds. The reference configuration reproduces the frozen exports exactly, at 0.000000 against both the transfer and the within-region references.

Region	Increment, full set	Increment without the two channels	Retained
Manavgat 2021	+0.067	+0.063	94 %
Bejís 2022	+0.056	+0.046	82 %
Muğla 2021	+0.116	+0.097	84 %
North Evia 2021	+0.153	+0.145	94 %
Montiferru 2021	+0.101	+0.105	103 %
Mean	+0.099	+0.091	92 %

The increment does not depend on them. It stays positive in every
1445 region and retains 82 % to 103 % of its size, rising slightly in one region. Mean
transfer is likewise unchanged, at 0.545 without the two channels against
0.541 with them and 0.537 for the baseline alone. The local skill this paper
reports is therefore not an artefact of a coordinate-smoothed surface.

This arm was computed independently here and reproduces the compan-
1450 ion paper’s figure for the same quantity, 82 % to 103 %, from a separately
written harness.

Appendix A.1. A(i). The four evaluations of Section 4.3, in full

Section 4.3 reports four evaluations of the same models as a ladder. The
per-split and per-scar detail is here, so that the ladder can be checked without
1455 leaving the manuscript.

Table A1. Within-region half-split, every split. Source and target
positive counts are given because they are unequal, which is the principal
limit on this arm: a straight cut does not produce two exchangeable halves.
Two splits are unusable because one half of Manavgat contains no burned
1460 cells.

Region	Axis	Direction	Source positives	Target positives	Thermal AUC	Baseline AUC
manavgat 2021	east-west	low to high	700	84	0.695	0.661
manavgat 2021	east-west	high to low	84	700	0.557	0.580
manavgat 2021	north-south	low to high	0	784	—	—
manavgat 2021	north-south	high to low	784	0	—	—
bejis 2022	east-west	low to high	640	460	0.750	0.784
bejis 2022	east-west	high to low	460	640	0.578	0.560
bejis 2022	north-south	low to high	357	743	0.713	0.662
bejis 2022	north-south	high to low	743	357	0.603	0.613
muqla 2021	east-west	low to high	1,449	1,462	0.448	0.378
muqla 2021	east-west	high to low	1,462	1,449	0.636	0.628
muqla 2021	north-south	low to high	774	2,137	0.533	0.503
muqla 2021	north-south	high to low	2,137	774	0.294	0.549
evia 2021 extended	east-west	low to high	2,052	612	0.589	0.506
evia 2021 extended	east-west	high to low	612	2,052	0.645	0.482
evia 2021 extended	north-south	low to high	2,564	100	0.649	0.513
evia 2021 extended	north-south	high to low	100	2,564	0.533	0.495
montiferru 2021	east-west	low to high	410	129	0.475	0.461
montiferru 2021	east-west	high to low	129	410	0.507	0.513
montiferru 2021	north-south	low to high	438	101	0.583	0.511
montiferru 2021	north-south	high to low	101	438	0.536	0.447

Table A2. Leave-one-scar-out at a 2 km buffer, every scar. Muqla and Montiferru are the only regions containing more than one burned component of at least 50 cells. Muqla is the only region where holding one out still leaves the source model properly trained across every arm; Montiferru’s two components are very unequal, so one of its arms retains 472 source positives and the other 97. Muqla’s four arms mean 0.579; the four arms in the other regions mean 0.525, and the pooled figure of 0.552 averages the two.

Region	Component	Source	Target positives	Target cells	AUC
		positives left			
1470 evia 2021 extended	1	11	2,653	3,059	0.465
manavgat 2021	1	88	696	1,135	0.592
montiferru 2021	1	97	442	758	0.584
montiferru 2021	5	472	67	195	0.458
mugla 2021	6	1,997	914	1,266	0.595
mugla 2021	1	2,173	738	1,244	0.561
mugla 2021	10	2,272	639	940	0.617
mugla 2021	8	2,363	548	954	0.542

The evaluation populations of the two arms are not comparable with each other or with the transfer targets. A held-out scar with its 2 km collar contains only fire-adjacent negatives, against a whole target region’s inclusion of its easy far field; the burned fractions, 34 to 87 % against 3.8 to 28.7 %, are a symptom of that rather than the cause, since ROC-AUC is invariant to class balance at fixed class-conditional distributions. The scar arm therefore asks for discrimination against the nearest and most similar negatives only, while a transfer arm includes the whole easy far field. Section 4.3 states the consequence: the last three rows of the ladder are not distinguishable by this design.

Table A3. The foreign-region arm, decomposed by source. Each held-out scar area is scored with a model fitted on each of the other four regions. Table 2’s row D is the mean over the **eight** scars that carry a row C, that is 32 of the 36 combinations below; the nine-scar mean quoted in this appendix is 0.559 against row D’s 0.555.

Target region	Scar	Mean over sources	Min	Max	Spread
Bejís 2022	1	0.589	0.521	0.631	0.110
North Evia 2021	1	0.590	0.374	0.704	0.329
Manavgat 2021	1	0.468	0.407	0.563	0.156
Montiferru 2021	1	0.504	0.457	0.546	0.089
Montiferru 2021	5	0.490	0.399	0.656	0.257
Muğla 2021	1	0.518	0.425	0.608	0.183
Muğla 2021	6	0.568	0.533	0.597	0.064
Muğla 2021	8	0.632	0.570	0.724	0.155
Muğla 2021	10	0.671	0.579	0.716	0.137

Over all 36 combinations here the mean is 0.559, the range 0.374 to 0.724, and ten fall below chance. Restricted to the 32 combinations behind row D the mean is 0.555 and the same ten fall below chance. The mean spread
1490 across the four sources for a single scar is 0.164 over the nine scars and 0.171 over the eight. Averaging over sources is what makes row D comparable with row C, which is fitted on one region; it is not a claim that the choice of foreign source is immaterial, and Section 4.3 states the distinction.

**Table A4. Prevalence is not the cause of the evaluation-area
1495 effect.** The same fitted model and the same out-of-fold predictions are scored three ways: on the whole region, on a random sample of region cells drawn at the scar area’s own burned fraction, and on the scar area. Twenty draws per scar, seed 42. Nine scars, since this control needs no leave-one-scar-out arm and Bejís therefore qualifies.

Target region	Scar	Burned fraction	A whole region	A'	B scar area
				prevalence- matched	
Manavgat 2021	1	0.61	0.797	0.796	0.589
Bejís 2022	1	0.66	0.824	0.825	0.575
Muğla 2021	6	0.72	0.777	0.772	0.564
Muğla 2021	1	0.59	0.777	0.779	0.654
Muğla 2021	10	0.68	0.777	0.779	0.673
Muğla 2021	8	0.57	0.777	0.782	0.761
North Evia 2021	1	0.87	0.864	0.866	0.745
Montiferru 2021	1	0.58	0.720	0.717	0.622
Montiferru 2021	5	0.34	0.720	0.723	0.461
Mean			0.782	0.782	0.627

A minus A', the effect of prevalence alone, is **−0.000** [**−0.003**, **+0.002**]. A' minus B, the effect of replacing the region's negatives with fire-adjacent ones, is **+0.155** [**+0.093**, **+0.217**]. The evaluation-area effect is entirely the negative pool.

Appendix A.2. A(j). The transfer-gap decomposition, in full

Section 4.5 states the two figures this table carries — at most 34 % of the gap recovered in the directions that started below chance, and seven directions with negative recovery. The table is kept here rather than in the body because Section 4.3 shows the within-region reference in its denominator is not matched to a transfer evaluation, and Section 4.4 shows its raw column is frame-dependent, so the fractions should be read as within-protocol quantities.

Table A5. Transfer-gap decomposition (four-AOI set, 12 directions). Within = target's within-region thermal AUC; best adapted = the

1515 better of z-score/CORAL; recovered fraction = (adapted - raw)/(within
- raw), signed and unclipped, with paired bootstrap CI (1000 replicates).
Montiferru directions are not part of this decomposition (per-pair absolute
decompositions exist without fraction CIs). The status column asks whether
the *adapted* value clears chance and uses the 2-cell adapted intervals of Ta-
1520 ble B9. The adapted arms were not recomputed at the coarser blocking of
Appendix A(c), which covers the raw arm and the paired delta only.

Direction	Within	Raw	Best adapted (method)	Recovered fraction [CI]	Status
Manavgat→Bejís	0.918	0.326	0.511 (CORAL)	+0.31 [+0.28, +0.34]	recovery, chance not excluded
Bejís→Manavgat	0.870	0.444	0.555 (CORAL)	+0.26 [+0.18, +0.34]	recovery above chance
Muğla→Manavgat	0.870	0.401	0.560 (CORAL)	+0.34 [+0.28, +0.40]	recovery above chance
Bejís→Evia	0.912	0.383	0.532 (z-score)	+0.28 [+0.23, +0.32]	recovery above chance
Evia→Bejís	0.918	0.448	0.549 (z-score)	+0.22 [+0.16, +0.27]	recovery above chance
Manavgat→Muğla	0.859	0.470	0.443 (CORAL)	−0.07 [−0.12, −0.03]	negative recovery
Muğla→Bejís	0.918	0.583	0.560 (CORAL)	−0.07 [−0.16, +0.01]	negative recovery
Manavgat→Evia	0.912	0.613	0.542 (z-score)	−0.23 [−0.32, −0.14]	negative recovery
Evia→Manavgat	0.870	0.686	0.527 (CORAL)	−0.86 [−1.18, −0.60]	negative recovery
Evia→Muğla	0.859	0.577	0.530 (CORAL)	−0.17 [−0.22, −0.11]	negative recovery
Muğla→Evia	0.912	0.653	0.563 (CORAL)	−0.35 [−0.43, −0.27]	negative recovery
Bejís→Muğla	0.859	0.618	0.518 (z-score)	−0.42 [−0.51, −0.34]	negative recovery

Appendix A.3. $A(k)$. The thermal sign, stratified

Section 4.4 reports that all five regions agree on a negative association
1525 between pre-fire surface temperature and burning, and that neither of the

two obvious confounders explains it. The per-region values are here. Signed AUC against **burned** within the 10 km collar; stratified columns pool within-stratum concordance over deciles of the named variable, weighting each stratum by its positive-negative pair count. Source `matched_frame_gap.csv`,
1530 recomputable by `paper/code/verify_matched_gap.py`.

Region	LST raw	LST within elevation	LST within NDVI	LST within distance	NDVI raw	NDVI within LST
Manavgat	0.386	0.402	0.440	0.505	0.621	0.542
Bejís	0.405	0.509	0.486	0.350	0.618	0.574
Muğla	0.332	0.363	0.404	0.367	0.652	0.547
Evia	0.286	0.327	0.279	0.319	0.663	0.380
Montiferru	0.376	0.392	0.368	0.485	0.582	0.405

Three readings follow. The LST sign survives stratification within elevation in four of five regions and within greenness in all five, despite $r(\text{LST}, \text{NDVI})$ reaching -0.92 , so it is neither a lapse-rate proxy nor an inverse-greenness proxy. It does *not* survive stratification within distance to the
1535 nearest burned cell in Manavgat or Montiferru, which are the two regions where the collar leaves the least residual gradient, so in those two the agreed sign is a weaker version of the same spatial effect the collar was introduced to
1540 remove. And the reciprocal adjustment runs one way only: NDVI reverses in two regions once temperature is held, while LST reverses in none once greenness is held.

Appendix A.4. A(l). The LST anomaly under the difference instrument

Section 4.4 reports that no reversal meets this paper’s strict criterion
1545 once evaluation frames are equalised, and that the LST anomaly nonetheless differs between regions on the weaker instrument Table B3’s note commits

the paper to. These are the four pairs that have opposite-sided point estimates and a difference interval excluding zero, all on `lst_anomaly_mean`. Signed AUC within the 10 km collar; the two regions are bootstrapped independently under the 10-cell spatial-block scheme and differenced, 1000 replicates, seed 42. Source `matched_frame_gap.csv`, recomputable by `paper/code/verify_matched_gap.py`.

Table A6. Between-region differences in the signed LST-anomaly association, 10 km collar.

	Pair	AUC A	AUC B	Difference	95 % CI
	Bejís vs Evia	0.392	0.584	−0.191	[−0.295, −0.083]
1555	Evia vs Montiferru	0.584	0.400	+0.184	[+0.012, +0.321]
	Manavgat vs Evia	0.462	0.584	−0.122	[−0.225, −0.020]
	Bejís vs Muğla	0.392	0.507	−0.115	[−0.219, −0.002]

Nine of the ninety feature-by-pair differences clear zero, against about 4.5 expected at nominal 5 % under the null, and no multiplicity correction is applied; four of the nine are this feature and three of those involve Evia. The nine features are effectively two to three dimensions (Section 4.4), so the ninety comparisons are not independent either. This is reported as a weaker result than a reversal, not as a restored one.

Appendix A.5. $A(m)$. The same-geography event pair

Section 4.10 states the result and its withdrawal; the design, the two structural asymmetries that have no analogue in the twenty-direction matrix, and the direction of the bias they impose are here.

This arm was designed to hold place fixed and vary only the fire, which would have separated regional concept shift from everything that differs be-

tween study areas. Muğla burned twice, in 2021 and again eleven months later, on the same grid and through the same processing chain, and the
1570 2022 arm is the 2021 population with the 2021 scar removed: 41,730 rows / 2,911 burned for 2021 against 38,790 rows / 331 burned for 2022. **Positive-carrying 5 km blocks: 70 for the 2021 arm and 11 for the 2022 arm,** and eleven is below the sixteen this design sets as its own floor, so the 2022 intervals are read as indicative exactly as the 20-cell row of Table 1 is. Full
1575 per-feature values are in Appendix B, Table B5.

On the frame as drawn, elevation reverses with bootstrap support. In 2021 higher ground burned preferentially, at 0.611 [0.532, 0.690]; in 2022 lower ground did, at 0.296 [0.230, 0.355]. The intervals are disjoint and the difference is -0.317 [-0.414 , -0.220]: the same predictor, region
1580 and grid, and an association pointing the opposite way in two fires eleven months apart. The four absolute thermal channels change side as well.

Section 4.4 has already withdrawn it. The 2022 arm is one compact scar inside the whole Muğla box, so 93.2 % of its cells lie beyond 10 km of any burned cell against 55.3 % for the 2021 arm — the most extreme far field in
1585 the cohort. Under the same 10 km collar the 2021 figure barely moves, 0.611 to 0.606 and still supported, while the 2022 figure moves from 0.297 to 0.565, onto the same side of 0.5 as 2021 with an interval covering chance. The arm shows the same artefact as the cross-region reversals rather than confirming them. We keep the section because the structural properties below have no
1590 analogue elsewhere in the paper, and because this arm is what motivated the frame test.

The two arms are not disjoint samples: they share 38,789 of the 2022

arm’s 38,790 cells, and elevation, slope and land cover are identical to the digit across all 73,098 grid cells, so only NDVI and the six thermal channels
1595 carry new information between them. And the removal is the target’s own positive class, so in the 2022-to-2021 direction not one target positive is in the source training population while 38,789 of 38,819 target negatives are, and membership of the source training set alone separates the 2021 target’s classes at ROC-AUC 0.9996. **We therefore cannot bound what that**
1600 **asymmetry does to a transfer estimate between the two arms.** The known part of the bias runs the safe way: the removed cells are high, with a median of 563 m against the population’s lower centre, so removing them strips high negatives from the 2022 arm and makes the observed reversal smaller rather than larger.

1605 *Appendix A.6. $A(n)$. The two interventions, in full*

Section 4.8 states both results; the four feature-set configurations, the per-region deltas and the pooled per-target shortfalls are here.

(a) **Pooled multi-region training** (Fig. C.6). At the point estimate, training on the pooled primary populations of the other four regions never
1610 beats the best single-source transfer for any target, with shortfalls of 0.02 to 0.22, and it stays 0.28 to 0.50 AUC below the within-region ceiling; for two targets it falls below the pairwise mean and below chance, though only Bejís is below chance with interval support, at 0.417 [0.369, 0.467]. Aggregation does not recover what single-source transfer loses.

1615 (b) **Removing the direction-reversing features.** The two predictors whose signed association reverses between regions **with bootstrap support on the frames as drawn** are `elevation_mean` and `lst_anomaly_mean`.

Section 4.4 withdraws that support under an equalised frame, so this selection rule is frame-dependent and the arm below measures what the removal costs under the original protocol, not a consequence of an established reversal. The per-region signed associations and the three pair-level reversals that meet that criterion are in Appendix B, Tables B2 and B3. Retraining without them costs **-0.081** of mean within-region AUC, supported in every region (per-region deltas -0.060, -0.130, -0.074, -0.063, -0.079, every interval entirely below zero), and changes mean transfer by **+0.014** [**-0.017**, **+0.045**], which spans zero.

Two qualifications belong with those numbers rather than after them. First, the debit is not the thermal block's. Dropping elevation alone accounts for -0.061 of it, from 0.888 to 0.827. Dropping the LST anomaly alone accounts for -0.013, from 0.888 to 0.875. Roughly three quarters of the cost is therefore the removal of a **baseline** terrain variable, not of a thermal channel. Second, both figures are post-selection. The two predictors were chosen because they reverse, using the same data on which the -0.081 and the +0.014 are then estimated. No correction for that selection is applied.

Configuration	Mean within-region AUC	Mean transfer AUC
full	0.888	0.541
drop elevation_mean	0.827	0.546
drop lst_anomaly_mean	0.875	0.544
drop both	0.807	0.556

Fig. C.7 shows the four configurations together. What this measures is a local cost with no compensating transfer gain. It does not measure an exchange, because the transfer side is a null on both arms. The thermal

block’s own contribution to transfer is +0.004, with an interval spanning
1640 zero (Section 4.5). Removing the reversing predictors returns +0.014, with
an interval spanning zero. Two nulls on the portability axis are not a price
paid.

Appendix A.7. A(o). The thermal sign and the anomaly difference, in full

Section 4.4 states both results; the per-region values, the reciprocal strat-
1645 ification and the difference-instrument argument are here.

A weaker instrument does support the anomaly result. Table
B3’s note commits this paper to a difference interval on the pair as the
sharper test, and it is applied here. Bootstrapping the two regions indepen-
dently under the collar and differencing, four pairs have opposite-sided point
1650 estimates **and** a difference interval excluding zero, all on `lst_anomaly_mean`
and three of the four involving Evia — the largest being Bejís against Evia at
−0.191 [−0.295, −0.083] (Appendix A(1), Table A6). The anomaly is the one
channel the frame-and-terrain mechanism of this section cannot explain, be-
ing decorrelated from elevation. Three caveats keep it weak and all three are
1655 stated rather than buried: nine of ninety feature-by-pair differences clear zero
against about 4.5 expected under the null with no multiplicity correction; the
nine features are effectively two to three dimensions, as the collinearity para-
graph below shows; and Evia’s own interval-support status turns on 0.003.
The honest statement is that no reversal meets this paper’s strict
1660 **criterion under the collar, and that the LST anomaly differs be-**
tween regions on the weaker difference instrument. We state it that
way because the alternative would be to apply a looser standard to the arm
that supersedes Table B3 than to Table B3 itself. Elevation under the collar

is above 0.5 in all five regions but individually supported in only two, Muğla
1665 at 0.606 [0.525, 0.685] and Evia at 0.648 [0.550, 0.740].

The sign the five regions agree on is not the one the dryness framing predicts. For LST the common direction is *below* 0.5 in every region, at 0.386, 0.405, 0.332, 0.286 and 0.376: a hotter pre-fire surface is associated with **less** burning, and the same holds for TVDI. It is not a lapse-
1670 rate artefact and it is not greenness acting through fuel load. On mutual adjustment the surviving channel is LST, not NDVI: holding NDVI, LST never reverses in any region, while holding LST, NDVI's own association reverses in Evia and Montiferru. Greenness is therefore not the mechanism. Two caveats bound what the sign is: in Manavgat and Montiferru it is a
1675 residual spatial gradient that disappears when distance to the nearest burned cell is stratified within the collar, and interval support is not uniform, LST being supported in four of five regions and TVDI in three, so "all five agree" is a statement about point estimates. What we can defend is that the absolute thermal channels behave here as **static land-surface descriptors** rather
1680 than as a dryness index, and that the two internally differenced channels carry no consistent cross-region direction at all. Compositing depth was not tested and remains an open alternative. Per-region stratifications are in Appendix A(k), and Section 5.2 states the consequence for the moisture-stress motivation of Section 1.2.

The same-geography arm of Section 4.10 is the most extreme case in the cohort, and it does not survive either. A fixed study area does not mean a fixed evaluation frame: its 2022 arm has **93.2 % of cells beyond 10 km of any burned cell, median 43.6 km**, a larger far field
1685

than any cross-region arm. Applying the same collar (`mugla_two_event_`
 1690 `collar.csv`; the full-frame values reproduce Table B5):

Muğla arm	share beyond 10 km	full frame	10 km collar
2021	55.3 %	0.611 [0.529, 0.692], supported	0.606 [0.525, 0.685], supported
2022	93.2 %	0.297 [0.229, 0.363], supported	0.565 [0.450, 0.677], not supported

The 2022 arm moves to the **same side of 0.5** as the 2021 arm and its interval covers chance, while the 2021 arm is nearly frame-invariant, which isolates the effect to the 2022 arm's far field rather than to the collar. **No**
 1695 **bootstrap-supported sign reversal survives anywhere in this paper once evaluation frames are equalised**, between regions or between two fires in one region.

Two points bound how many independent reversals could have been counted in the first place. The channels whose reversal vanishes are the ones partly
 1700 measuring terrain: current LST correlates with elevation at -0.695 to -0.125 across the regions and TVDI at -0.722 to -0.298 , so their proxy strength varies fivefold and a frame that shifts the elevation distribution shifts them with it. And within the collar `fused_lst_mean` correlates with `current_lst_mean` at 0.99 to 1.00, `downscaled_lst_mean` at 0.97 to 0.99 and `current_tvdi_mean`
 1705 `tvdi_mean` at 0.87 to 0.98, so the four absolute channels are close to one axis and the two differenced channels correlate at 0.64 to 0.94: the "five of nine directions reverse" count of Section 4.7 counts features, not independent quantities, and **in effective dimensions it is closer to two**.

Two scope statements belong with the table. The full-frame values here
 1710 are not numerically the published ones — $\rho = +0.86$ over fourteen directions
 against the $+0.84$ over sixteen in Section 4.6 — because the support test
 is itself bootstrap-dependent and at this setting three region pairs carry no
 jointly supported feature rather than two; the argument rests on before-and-
 after under identical settings, not on the published figure. And **only these**
 1715 **diagnostics were recomputed on the collar**: the other eighteen failed
 on the frames as drawn and were not rerun, so their correlations against the
 equalised transfer vector are unknown rather than shown to be null.

The thermal block’s paired contribution to transfer, with its in-
terval. The two matrices were differenced direction by direction. The mean
 1720 is **+0.004**, but the mean is not the informative statistic here. The individ-
 ual paired contributions span **-0.148 to +0.133**, with **twelve positive**
and eight negative (`baseline_vs_thermal_transfer.csv`). The spread
 is thirty times the mean, and the sign is not a property of the block but of
 the pair it is asked to cross: the same six predictors that add $+0.133$ in one
 1725 direction subtract 0.148 in another. A mean near zero here records cancel-
 lation, not consistent absence of effect. The directions are not independent,
 because each region appears in eight of the twenty. The interval therefore
 depends on what is treated as the resampling unit. All four units the design
 permits give the same answer:

	Resampling unit	n	95 % interval on the mean paired contribution
1730	Directions, naive	20	$[-0.027, +0.034]$
	Unordered pairs, cluster bootstrap	10	$[-0.028, +0.036]$
	Unordered pairs, t on pair means	10	$[-0.034, +0.042]$
	Regions, leave-one-out jackknife	5	$[-0.037, +0.046]$

Every interval spans zero, and the point estimate is a small fraction of each width. The per-region jackknife shows how little the mean is anchored. Holding out Manavgat, Bejís, Muğla, Evia and Montiferru in turn gives +0.0148, +0.0021, +0.0065, **-0.0081** and +0.0060. **Dropping Evia alone**
1735 **reverses the sign of the headline.** None of these four units propagates within-direction sampling variability. They resample between directions only.

In precision terms the transfer is worse than the ROC figures suggest. ROC-AUC is the metric used throughout this paper, for comparability with the literature, but a susceptibility surface is used as a ranked
1740 area budget, so precision-recall is the operational quantity. Across the twenty directions the thermal model’s PR-AUC averages **0.156 against a no-skill baseline of 0.136.** The mean of the twenty per-direction lifts is 1.16; the ratio of the two means just quoted is 1.146. **Six of the twenty fall below their own no-skill baseline at the point estimate, and five of those**
1745 **six have intervals entirely below it.** The exception is Bejís to Manavgat, at 0.034 $[0.029, 0.042]$ against a baseline of 0.038, whose interval covers the baseline and which is therefore a point-estimate case only. Only one direction, Evia to Manavgat, exceeds twice its baseline, at 0.094 against 0.038. A transferred model therefore ranks burned cells about a sixth better than

1750 random on average, and worse than random in five directions with interval support and a sixth at the point. This is a sharper statement than the ROC means support and it should be the one a practitioner reads.

The within-region increment survives the same correction, and is reported here because Section 4.2 establishes it on the frames this
1755 **section calls incomparable.** Running the baseline arm on the collar as well (`collar_increment_and_cosine.csv`, `paper/code/verify_collar_increment.py`), the thermal increment at 5 km blocking is +0.041, +0.030, +0.091, +0.134 and +0.090 across the five regions, **positive in all five**, with a mean of +0.077 against +0.086 on the frame as drawn. It does erode as the frame
1760 tightens further: at a 5 km collar the mean halves to +0.041, still positive in all five. These are point estimates; the intervals in Table 1 are computed on the frame as drawn. The paper’s one surviving predictor-level positive claim therefore holds on the frame it argues is the correct one.

The five areas of interest are not comparable frames. Each region
1765 is a rectangle drawn around a fire, and the rectangles differ by an order of magnitude in how much unburnt far field they enclose.

The share of modelled cells lying beyond 10 km of any burned cell is 60.1 % in Manavgat, 63.1 % in Bejís, 55.3 % in Muğla, 43.7 % in Evia and **2.1 %** in Montiferru, with median distances of 13.4, 13.5, 11.3, 8.0 and 2.7
1770 km (Appendix B, Table B6). Montiferru’s frame is fire-scale; Manavgat’s and Bejís’s are roughly three-fifths far field. The far field is not a neutral addition. In Manavgat the median elevation of modelled cells rises from 472 m within 5 km of the fire to 955 m at 10 to 20 km and 1,273 m at 20 to 50 km, against 512 m for the burned cells themselves, in Taurus terrain that no

1775 plausible spread model would place at risk.

Appendix B. Appendix B. All twenty transferability diagnostics

Section 4.6 summarises this table by family. The full ranking is given here because the negative result is the point: a reader should be able to see every candidate that was tried, not only the families' best members, and to
1780 check that no diagnostic was dropped after it failed.

Table B1. All transferability diagnostics versus raw thermal transfer (20 ordered directions). Spearman ρ with pair-based bootstrap 95 % CIs. Exp. = expected sign. Rows with $n = 12$ exist only for the four-AOI subset, because those diagnostics were never produced for Montiferru.
1785 The supported-features conditional rows use the 16 directions (8 pairs) with at least one CI-supported feature.

Diagnostic	Family	Exp.	n	dir	Spearman ρ [95% CI]	CI excludes 0
Agreement fraction, supported features	P(y x) conditional	+	16		+0.84 [+0.58, +0.88]	yes
Cosine, supported features	P(y x) conditional	+	16		+0.81 [+0.33, +0.88]	yes
Cosine, all 9 features	P(y x) conditional	+	20		+0.50 [−0.17, +0.83]	no
Vector	P(y x)	+	20		+0.27 [−0.36, +0.77]	no
Spearman, all 9	conditional					
Agreement count, all 9	P(y x) conditional	+	20		+0.18 [−0.40, +0.72]	no
Schoener’s D, 1-D mean	P(x y=1) niche	+	20		+0.24 [−0.45, +0.74]	no
Warren’s I, 1-D mean	P(x y=1) niche	+	20		+0.22 [−0.42, +0.73]	no
Schoener’s D, PCA-2D	P(x y=1) niche	+	20		+0.10 [−0.51, +0.68]	no
Warren’s I, PCA-2D	P(x y=1) niche	+	20		−0.07 [−0.66, +0.49]	no
Mahalanobis, burned centroids	P(x y=1) niche	−	20		−0.23 [−0.75, +0.44]	no
Domain-classifier AUC	P(x) marginal	−	20		−0.32 [−0.78, +0.33]	no
Predictor-space mean dissimilarity	P(x) marginal	−	12		−0.10 [−0.54, +0.43]	no
Predictor-space p95 dissimilarity	P(x) marginal	−	88	12	−0.08 [−0.59, +0.49]	no
Fraction inside weighted AoA	P(x) marginal	+	12		+0.22 [−0.48, +0.59]	no
Fraction inside unweighted	P(x) marginal	+	12		+0.08 [−0.89, +0.63]	no

Table note. A null row means the diagnostic was **not shown to order transfer** on this design, not that it was shown incapable of ordering it.

1790 With ten effective region pairs the power is low, and the intervals are wide enough to admit moderate true correlations in either direction. The last row is retained rather than deleted because it was computed: on two directions the statistic has no meaningful value, and reporting that is more honest than dropping the variant.

1795 **Reading the two rows that clear zero.** Both are supported-feature variants, where the predictor subset is chosen by whether two regions' bootstrap intervals happen to be disjoint. That is a data-dependent selection made on the same data, uncorrected. Their all-nine-feature counterparts are in the table and both span zero. The result lives in the selection step, and
1800 Section 4.6 says so.

Equal-sample check. The families sit on unequal samples: marginal, applicability, climatic and geographic rows on twelve directions, the supported-conditional rows on sixteen, the rest on twenty. Recomputing every row on the common twelve directions reproduces the published values to 4.8×10^{-5}
1805 and leaves the ordering unchanged — the conditional rows still lead at +0.87 [+0.65, +0.88] and +0.85 [+0.43, +0.88], every marginal row still spans zero. Source: `paper/diagnostics_common_subset.md`.

Appendix B.1. B2. The signed univariate associations the reversal claim rests on

1810 Section 4.8b drops two predictors because their signed association with burning reverses between regions with bootstrap support on the frames as drawn. Section 4.4 shows that both supported elevation reversals disappear

once the frames are equalised, so this table is the evidence for the feature-removal arm and for the narrowed mechanism claim inside Contribution 1,
1815 not for a general reversal mechanism. It was computed for the frozen analysis and is reproduced here because the claim is otherwise asserted rather than shown.

Table B2. Signed univariate AUC of each predictor against burned, by region. Primary natural-vegetation population; 10-cell (~ 5
1820 km) spatial-block bootstrap, 1000 replicates, seed 42. The AUC is never folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a value below 0.5 means lower values rank burned and is a direction rather than weakness. **Bold** marks a region whose own interval excludes 0.5.

Feature	Manavgat	Bejís	Muğla	Evia	Montiferru
elevation_	0.374 [0.289,	0.643 [0.558,	0.611 [0.532,	0.541 [0.448,	0.584 [0.395,
mean	0.471]	0.729]	0.690]	0.626]	0.762]
slope_mean	0.531 [0.423,	0.521 [0.439,	0.637 [0.582,	0.487 [0.418,	0.652 [0.506,
	0.642]	0.605]	0.686]	0.554]	0.771]
ndvi_mean	0.636 [0.587,	0.559 [0.497,	0.662 [0.616,	0.639 [0.575,	0.586 [0.450,
	0.676]	0.619]	0.704]	0.701]	0.704]
lst_anomaly_	0.482 [0.428,	0.418 [0.364,	0.485 [0.395,	0.640 [0.567,	0.395 [0.285,
mean	0.530]	0.480]	0.566]	0.710]	0.535]
current_lst_	0.538 [0.452,	0.477 [0.401,	0.325 [0.271,	0.377 [0.301,	0.370 [0.248,
mean	0.621]	0.547]	0.382]	0.456]	0.513]
current_	0.552 [0.460,	0.517 [0.429,	0.336 [0.275,	0.362 [0.285,	0.356 [0.233,
tvdi_mean	0.641]	0.595]	0.398]	0.442]	0.499]
tvdi_	0.449 [0.391,	0.512 [0.443,	0.490 [0.396,	0.519 [0.444,	0.378 [0.282,
difference_	0.505]	0.583]	0.575]	0.589]	0.497]
mean					
downscaled_	0.552 [0.466,	0.484 [0.400,	0.307 [0.253,	0.377 [0.297,	0.365 [0.240,
lst_mean	0.637]	0.560]	0.366]	0.459]	0.511]
fused_lst_	0.540 [0.454,	0.481 [0.404,	0.325 [0.272,	0.376 [0.300,	0.370 [0.248,
mean	0.622]	0.551]	0.383]	0.456]	0.513]

1825 **What counts as a reversal.** A pair of regions is called a reversal

only when their signed associations point to opposite sides of 0.5 **and each region’s own interval excludes 0.5**. That is stricter than requiring the two regions’ intervals to be disjoint, and the difference matters: for `current_1st_mean` between Manavgat and Muğla the two intervals are disjoint, but
1830 Manavgat’s own interval, [0.452, 0.621], includes 0.5, so Manavgat has no established direction to reverse from. That pair is a point reversal, not a supported one.

Table B3. The cross-region reversals that meet the stricter criterion. Difference intervals are from the same paired bootstrap.

Feature	Region A	AUC	Region B	AUC	Difference [95 % CI]
<code>elevation_mean</code>	Manavgat	0.374	Bejís	0.643	+0.269 [+0.138, +0.397]
<code>elevation_mean</code>	Manavgat	0.374	Muğla	0.611	+0.235 [+0.102, +0.360]
<code>1st_anomaly_mean</code>	Bejís	0.418	Evia	0.640	+0.221 [+0.123, +0.313]

Three pair-level reversals across **two** features, which is why Section 3.12 removes exactly those two. Twenty-nine further pairs reverse at the point estimate only, spread across eight of the nine features, and they are not counted. The conservative criterion costs the paper findings rather than
1840 manufacturing them: a difference interval on the pair, which is the instrument Section 4.10 uses, would support more reversals than the three listed here.

Appendix B.2. B4. The transfer matrix in precision-recall terms

ROC-AUC is reported throughout the main text for comparability with the susceptibility literature. A susceptibility surface is used as a ranked
1845 area budget, so precision-recall is the operational quantity, and at target prevalences of 3.8 to 28.7 % the two can differ sharply. Read from the same frozen step9b exports as Table B9.

Table B4. Thermal transfer, PR-AUC against the no-skill baseline. The baseline is the target's burned prevalence. Lift is PR-AUC divided
1850 by that baseline; a lift of 1 is no better than random ranking. Ordered by lift.

Direction	ROC-AUC	PR-AUC	No-skill	Lift
Evia $\rightarrow$ Manavgat	0.686	0.094	0.038	2.45
Bejís $\rightarrow$ Montiferru	0.594	0.289	0.212	1.36
Evia $\rightarrow$ Montiferru	0.647	0.283	0.212	1.34
Bejís $\rightarrow$ Muğla	0.619	0.093	0.070	1.33
Muğla $\rightarrow$ Evia	0.653	0.379	0.287	1.32
Montiferru $\rightarrow$ Bejís	0.548	0.093	0.072	1.28
Montiferru $\rightarrow$ Muğla	0.619	0.089	0.070	1.27
Evia $\rightarrow$ Muğla	0.577	0.086	0.070	1.23
Muğla $\rightarrow$ Bejís	0.583	0.088	0.072	1.22
Manavgat $\rightarrow$ Evia	0.613	0.343	0.287	1.20
Manavgat $\rightarrow$ Montiferru	0.533	0.243	0.212	1.15
Montiferru $\rightarrow$ Manavgat	0.567	0.043	0.038	1.11
Montiferru $\rightarrow$ Evia	0.586	0.317	0.287	1.11
Muğla $\rightarrow$ Montiferru	0.531	0.214	0.212	1.01
Bejís $\rightarrow$ Manavgat	0.444	0.034	0.038	0.90
Manavgat $\rightarrow$ Muğla	0.470	0.063	0.070	0.90
Evia $\rightarrow$ Bejís	0.448	0.059	0.072	0.82
Bejís $\rightarrow$ Evia	0.383	0.226	0.287	0.79
Muğla $\rightarrow$ Manavgat	0.401	0.029	0.038	0.75
Manavgat $\rightarrow$ Bejís	0.326	0.049	0.072	0.67
Mean	0.541	0.156	0.136	1.16

Six directions fall below their own no-skill baseline, and only one exceeds twice it. The six are the same six that are below chance on ROC-AUC, which is what a reversed ranking predicts in either metric. Section 4.4 shows that this count is largely a property of the evaluation frames rather than of

a reversed predictor-burning relationship.

Appendix B.3. B5. The same-geography event pair, in full

Section 4.10 reports this arm and Section 4.4 withdraws its reversal as
1860 a frame artefact. The per-feature values are kept here because the arm is
what motivated the frame test, and because its structural asymmetries have
no analogue in the twenty-direction matrix.

**Table B5. Signed univariate feature-burned AUC, Muğla 2021
versus 2022.** Raw AUC against burned, never folded to $\max(\text{AUC}, 1 -$
1865 $\text{AUC})$; 10-cell (≈ 5 km) spatial-block bootstrap, 1,000 replicates, seed 42
(Section 3.15). Analysis population 41,730 rows / 2,911 burned (2021) and
38,790 rows / 331 burned (2022). **Positive-carrying 5 km blocks: 70 for
the 2021 arm and 11 for the 2022 arm.** Table 1’s note sets sixteen as
the floor this design supports at that blocking, so the 2022 intervals here fall
1870 below the paper’s own standard and are read as indicative, exactly as the 20-
cell row of Table 1 is. The 2022 arm is additionally a single compact scar, so
its eleven blocks are contiguous. Read from `paper/step9g_raw/.../mugla_`
`2021__mugla_2022_event_relative/step9g_direction_reversal_table.`
`csv`.

Feature	2021 AUC [95 % CI]	2022 AUC [95 % CI]	Reversal
elevation_mean	0.611 [0.532, 0.690]	0.296 [0.230, 0.355]	bootstrap- supported
current_lst_mean	0.325 [0.271, 0.382]	0.515 [0.434, 0.580]	point only
current_tvdi_mean	0.336 [0.275, 0.398]	0.594 [0.475, 0.674]	point only
downscaled_lst_mean	0.307 [0.253, 0.366]	0.508 [0.435, 0.571]	point only
fused_lst_mean	0.325 [0.272, 0.383]	0.519 [0.436, 0.583]	point only
ndvi_mean	0.662 [0.616, 0.704]	0.707 [0.624, 0.777]	none
slope_mean	0.637 [0.582, 0.686]	0.558 [0.468, 0.634]	none
lst_anomaly_mean	0.485 [0.395, 0.566]	0.380 [0.249, 0.502]	none
tvdi_difference_mean	0.490 [0.396, 0.575]	0.397 [0.265, 0.514]	none

Appendix B.4. B6. The evaluation frames, and the signed associations they produce

Section 4.4 states these results; the per-region values are here.

Table B6. Evaluation-frame geometry of the five study regions.

Primary natural-vegetation population. Distance is Euclidean to the nearest burned cell on the 500 m grid, at 0.45 km per cell. Source `aoi_frame_auc.csv`; recomputable by `paper/code/verify_aoi_frame.py`.

Region	cells	burned	median distance to burned	share beyond 10 km
Manavgat	20,511	784	13.4 km	60.1 %
Bejís	15,190	1,100	13.5 km	63.1 %
Muğla	41,730	2,911	11.3 km	55.3 %
Evia	9,298	2,664	8.0 km	43.7 %
Montiferru	2,544	539	2.7 km	2.1 %

Table B7. Signed univariate AUC, frame as drawn against a 10 km collar. Point estimates; the intervals that decide the reversal question are given in the text below and in `collar_frame_bootstrap.csv` (10-cell blocks, 1000 replicates, seed 42). Signed and never folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a value below 0.5 is a direction, not weakness. The collar drops no burned cells in any region.

Signed univariate AUC	Manavgat	Bejís	Muğla	Evia	Montiferru	straddles 0.5
elevation, full frame (Table B2)	0.374	0.643	0.611	0.541	0.584	yes
elevation, 10 km collar	0.561	0.614	0.606	0.648	0.581	no
current LST, full frame	0.538	0.477	0.325	0.377	0.370	yes
current LST, 10 km collar	0.386	0.405	0.332	0.286	0.376	no
current TVDI, full frame	0.552	0.517	0.336	0.362	0.356	yes
current TVDI, 10 km collar	0.392	0.454	0.342	0.250	0.361	no

Table B8. Region summary: populations and gate outcomes. Counts from each region’s Step 8A dataset statistics; gate fractions from each region’s burned-landcover gate output. TSG = the primary natural-vegetation population. The TSG columns use the canonical modelled pop-
1895 ulation, `burnable_tree_shrub_grass` and `valid_for_modeling == True`, which is the population every model in this paper was fitted and scored on.

Region	Total cells	Valid cells	Burned	Prevalence (all valid)	TSG cells	Burned in TSG	TSG prevalence	Burned natural- veg frac- tion	Grade verdict
Manavgat 2021	24,450	24,087	796	0.033	20,511	784	0.038	0.984	pass
Bejís 2022	15,759	15,759	1,103	0.070	15,190	1,100	0.072	0.991	pass
Mugla 2021	73,098	73,045	3,026	0.041	41,730	2,911	0.070	0.958	pass
North Evia 2021 (ex- tended)	22,925	22,906	2,788	0.122	9,298	2,664	0.287	0.945	pass
Monte Lezvi 2021	3,234	3,173	697	0.220	2,544	539	0.212	0.723	pass

Table B9. Cross-region transfer matrix, thermal model, TSG population. Target ROC-AUC with 2-cell spatial-block bootstrap 95% CIs

(1000 replicates). CORAL is applied after region-wise z-scoring ($\lambda = 10^{-5}$).

Direction	Raw	Region-wise z-score	CORAL
Manavgat→Bejís	0.326 [0.305, 0.349]	0.477 [0.451, 0.502]	0.511 [0.484, 0.534]
Bejís→Manavgat	0.444 [0.408, 0.480]	0.457 [0.420, 0.497]	0.555 [0.528, 0.583]
Manavgat→Muğla	0.470 [0.451, 0.488]	0.431 [0.411, 0.449]	0.443 [0.423, 0.461]
Muğla→Manavgat	0.401 [0.378, 0.426]	0.559 [0.531, 0.587]	0.560 [0.535, 0.587]
Manavgat→Evia	0.613 [0.593, 0.631]	0.542 [0.520, 0.565]	0.539 [0.518, 0.561]
Evia→Manavgat	0.686 [0.653, 0.716]	0.516 [0.489, 0.544]	0.527 [0.500, 0.553]
Bejís→Muğla	0.618 [0.601, 0.635]	0.518 [0.501, 0.535]	0.507 [0.489, 0.524]
Muğla→Bejís	0.583 [0.561, 0.607]	0.535 [0.512, 0.557]	0.560 [0.538, 0.581]
Bejís→Evia	0.383 [0.363, 0.402]	0.532 [0.509, 0.551]	0.499 [0.479, 0.518]
Evia→Bejís	0.448 [0.426, 0.470]	0.549 [0.524, 0.575]	0.549 [0.525, 0.573]
Muğla→Evia	0.653 [0.636, 0.671]	0.561 [0.543, 0.580]	0.563 [0.545, 0.582]
Evia→Muğla	0.577 [0.560, 0.593]	0.501 [0.485, 0.518]	0.530 [0.515, 0.546]
Montiferru→Manavgat	0.567 [0.539, 0.594]	0.573 [0.538, 0.609]	0.606 [0.574, 0.639]
Manavgat→Montiferru	0.533 [0.488, 0.580]	0.586 [0.540, 0.629]	0.592 [0.550, 0.631]
Montiferru→Bejís	0.548 [0.521, 0.578]	0.574 [0.552, 0.596]	0.569 [0.548, 0.591]
Bejís→Montiferru	0.594 [0.560, 0.631]	0.550 [0.500, 0.601]	0.574 [0.530, 0.621]
Montiferru→Muğla	0.619 [0.604, 0.634]	0.576 [0.562, 0.589]	0.565 [0.550, 0.579]
Muğla→Montiferru	0.531 [0.495, 0.568]	0.587 [0.549, 0.624]	0.584 [0.547, 0.623]
Montiferru→Evia	0.586 [0.565, 0.606]	0.630 [0.611, 0.649]	0.624 [0.605, 0.641]
Evia→Montiferru	0.647 [0.608, 0.682]	0.568 [0.528, 0.609]	0.581 [0.539, 0.623]

Table B10. The most and least environmentally similar pairs, on both frames. Schoener’s D is computed over burned cells only and is therefore collar-invariant. Transfer values are the two ordered directions of each pair; ranks are out of the twenty directions on the equalised frame. As-drawn transfer is read from Table B9, collar transfer from `aoi_frame_transfer.csv`.

	Manavgat–Muğla	Bejís–Montiferru
Schoener’s D , mean 1-D	0.826 (highest)	0.479 (lowest)
per-feature D	0.77 to 0.89	0.23 to 0.77
transfer, frames as drawn	0.470, 0.401	0.594, 0.548
transfer, 10 km collar	0.551, 0.510	0.669, 0.624
rank of 20 on the collar	5th, 2nd from bottom	15th, 11th
target cells inside the AoA	0.875, 0.531	—

1910 Appendix C. Appendix C. Protocol detail

These protocols are given here in full rather than in Methods, because each is a specification a reader needs only when checking the corresponding result, and none is needed to follow the argument. Sections 3.2, 3.3, 3.11 and 3.15 state in summary what each does.

1915 Appendix C.1. C.1 Burned-area label, the reconstructed analysis grid, and the admissibility gate

Labels come from MODIS MCD64A1 Collection 6.1 (MODIS/061/MCD64A1) retrieved through Google Earth Engine [11], whose omission and commission characteristics [4] bound every model here. The product is exported onto
1920 the 30 m EPSG:4326 reference grid used by the rest of the pipeline, which duplicates each native ~ 500 m observation across the 30 m pixels beneath it.

The analysis grid is therefore **reconstructed rather than native**: the 30 m grid is partitioned into non-overlapping blocks of $\text{round}(500 / 30) =$
1925 17×17 pixels, giving a nominal cell edge of 510 m, and each cell is identified by its integer block indices `row_500m` and `col_500m`. This approximates the

MODIS sinusoidal cell rather than reproducing it, being anchored to the Landsat/EPSG:4326 reference grid, and marginal blocks are truncated. The cell is square in degrees but not on the ground, about 510 m north to south and 390 to 407 m east to west, so its area is 0.199 to 0.208 km² against a MODIS cell's 0.250 km²; every block-size label in this paper is therefore the north-south dimension. The companion paper develops what follows from that.

A cell's representative burn date is the **mode** of the positive sub-pixel day-of-year values in the block, tested against the region's label window. Because the exported raster carries no out-of-window positives, in this dataset a single in-window positive sub-pixel makes the cell burned. The fraction of positive sub-pixels agreeing with the modal date is recorded as **burn_date_pixel_agreement_fraction** but no agreement threshold is imposed. The label never affects a cell's eligibility for modelling: unburned, all-no-data and out-of-window cells all remain as the negative class.

Two safeguards are recorded rather than assumed. Cells that burned inside a region's own predictor window are removed from its analysis universe rather than counted as unburned, which ran for Muğla, Evia and Montiferru, is recorded as not run for Manavgat and has no recorded status for Bejís. Burning in earlier years is screened for no region in the five-region cohort; the only historical exclusion in the study removes the 2021 Muğla scar from the 2022 event-relative experiment of Section 3.15.

Before any modelling each region passes a gate that answers one question: of the cells labelled burned, what fraction is dominated by natural vegetation? A region is admitted as a wildfire candidate when that fraction

reaches 0.50 and at least 30 burned cells are present, and is rejected as a cropland-dominated control when the cropland fraction reaches 0.50 instead. The gate uses ESA WorldCover classes aggregated to the same reconstructed
1955 cells. Its purpose is to separate burned area produced by natural-fuel combustion from burned area produced by post-harvest stubble burning, which MCD64A1 does not distinguish. Verdicts are reported in Section 4.1.

Appendix C.2. C.2 Transferability diagnostics versus transfer

Twenty candidate diagnostics from five families are each rank-correlated
1960 against the same target quantity, the raw thermal transfer AUC over the twenty ordered directions, under one common pair-based bootstrap. The families are marginal predictor-distribution measures $P(x)$, burned-niche overlap $P(x|y=1)$, fire-regime label-pattern structure $P(y)$, and conditional feature-response direction $P(y|x)$.

1965 The **marginal** family includes area-of-applicability-style dissimilarity in scaled, importance-weighted predictor space [23, 24, 18], climatic and geographic distance, and a learned domain classifier trained to separate source from target cells. The **niche-overlap** family includes Schoener’s D [36] and Warren’s I [44] over the burned cells of each region. The **regime** family in-
1970 cludes burned-patch component counts and effective component counts. The **conditional** family is the sign-agreement index: the fraction of predictors whose signed univariate association points the same way in both regions, computed over all predictors and over the subset whose associations are individually interval-supported.

1975 Two properties of this design are stated in advance because they bound what it can show. The candidate set was fixed in a project analysis log before

the correlations were computed; that log is not a formal pre-registration and no independent timestamped registration exists. And with five regions the effective sample is ten unordered pairs, so nineteen computed variants are
1980 evaluated at that power with no family-wise error control claimed; where a diagnostic clears an interval test but not a Bonferroni threshold, both are reported.

Appendix C.3. C.3 Same-geography event-to-event comparison (Muğla 2021 versus 2022)

1985 Muğla is the one region for which a second fire event is analysed on the identical AOI and analysis grid, which allows the direction of feature-label associations to be compared with geography held fixed. The 2022 experiment is registered with windows anchored to its own event: a 58-day predictor window closing the day before ignition and a 49-day label window opening
1990 on it, matching the 2021 durations exactly.

This is not a clean temporal-transfer design and is not presented as one. The 2022 event ignites about six weeks earlier in the season than the 2021 event, so calendar year and seasonal phase are confounded and no difference can be attributed to the year alone. The registered claim is same-
1995 geography event-to-event. The pair is deliberately kept out of the twenty-direction matrix, and that exclusion is enforced in released code rather than merely asserted here.

Two further properties bound the comparison. A historical-burn exclusion applies to the 2022 arm alone, masking every cell that burned in 2021 so the
2000 previous year’s scar does not enter the following year’s analysis: it removes 3,073 cells, of which 2,941 belong to the primary population, taking it from

41,730 rows to 38,790, a drop of 7.0 %. The two arms therefore share 38,789 of 38,790 cells and three byte-identical static predictors, and in the 2022-to-2021 direction every target positive lies outside the source training population. Section 4.10 states what follows for how these two numbers may be read.

Table C1. Study regions, areas of interest and temporal windows. Bounding boxes are in EPSG:4326, as registered. Window lengths in brackets are inclusive day counts. The baseline years are the four window-symmetric years preceding each predictor window.

Region	Bounding box (lon min, lat min, lon max, lat max)	Predictor window	Label window	Baseline years
Manavgat 2021 (Türkiye)	31.05, 36.72, 31.85, 37.35	2021-06-01 to 2021-07-27 (57 d)	2021-07-28 to 2021-08-31 (35 d)	2017, 2018, 2019, 2020
Bejís 2022 (Spain)	-1.05, 39.68, -0.35, 40.15	2022-06-15 to 2022-08-14 (61 d)	2022-08-15 to 2022-09-30 (47 d)	2018, 2019, 2020, 2021
Muğla 2021 (Türkiye)	27.1, 36.6, 28.9, 37.45	2021-06-01 to 2021-07-28 (58 d)	2021-07-29 to 2021-09-15 (49 d)	2017, 2018, 2019, 2020
North Evia 2021 (Greece)	23.05, 38.55, 23.85, 39.15	2021-06-05 to 2021-08-02 (59 d)	2021-08-03 to 2021-09-30 (59 d)	2017, 2018, 2019, 2020
Montiferru 2021 (Italy)	8.45, 40.05, 8.75, 40.27	2021-05-25 to 2021-07-23 (60 d)	2021-07-24 to 2021-08-31 (39 d)	2017, 2018, 2019, 2020

Appendix C.4. Predictor variables

Ten predictors are used, four baseline and six thermal, all summarised per cell over the predictor window, through the chain shown in Fig. C.2. All optical and thermal predictors come from Landsat 8 Collection 2 Level-2 (LANDSAT/LC08/C02/T1_L2), quality-screened per pixel from the QA_PIXEL band with the water bit deliberately preserved; the coarse-resolution thermal input is MODIS/061/MOD11A1.

- **NDVI**, the predictor-window median of Landsat surface reflectance, is the baseline’s one time-varying member.
- 2020 • **Elevation** and **slope** come from the Copernicus DEM GLO-30.
- **Land cover** is ESA WorldCover v200 [48], entering as the dominant class code.
- **Current LST**, the predictor-window median of Landsat surface temperature in °C.
- 2025 • **LST anomaly**, a z-score of the current-window LST median against the four baseline years, set to no-data where the baseline standard deviation is below 1.0 °C or observations are too few.
- **TVDI** and **TVDI difference**. The Temperature-Vegetation Dryness Index [35] is computed in the LST-NDVI feature space, with wet and dry edges taken as the 2nd and 98th LST percentiles within each of twenty NDVI bins and the index clamped to [0, 1]. The edges are percentiles of the LST values a given scene contains, so they are fitted per AOI and per window, and a TVDI of 0.5 denotes a different physical moisture state in each region. The companion paper reports what that scene dependence does and does not explain. **TVDI difference** is the raw anomaly of that index against the same four window-symmetric baseline years used for the LST anomaly: `tvdi_difference = current_tvdi - mean(baseline_tvdi)`, in index units rather than standard deviations, which is why it is reported alongside the z-scored channel rather than in place of it.
- 2030
- 2035
- 2040

• **Downscaled LST and fused LST.** A MODIS-to-Landsat downscal-
 ing model is trained on the predictor window and applied to the full 30
 m grid. The fused product equals observed Landsat LST wherever that
 is valid, and the downscaled surface only where it is not. Gap-filling
 therefore never replaces or blends a valid observation. The gap-filled
 share is 0.11 % to 9.70 % by region. The downscaler’s own inputs in-
 clude coordinates, which is the one route by which a coordinate-derived
 surface re-enters a feature set from which Section Appendix C.6 ex-
 cludes coordinates. Appendix A(g) reports the increment without these
 two channels.

Appendix C.5. C.4 Limitations, in full

Section 5.9 states each limitation; the elaborations are here.

(i) **No meteorological covariates** enter the models, so we cannot say
 how local skill and portability behave for a mixed thermal-plus-weather pre-
 dictor set.

(ii) **The same-geography comparison covers one region only**, and
 even there year and seasonal phase are confounded, a confound that cannot
 be resolved in this study area for the reason given in Section 5.2. Its
 burned cells also leave the thermal reversals unresolved at interval level,
 and the pair holds place fixed but not population. Those two arms are
 additionally the only transfer directions here computed by us rather than
 read from the pipeline author’s frozen export, with his unmodified code and
 the same pinned environment.

(iii) **All labels derive from a single burned-area product**, MCD64A1
 [10], whose omission and commission characteristics [4] bound every model

evaluated here. No second burned-area product covers 2021 and 2022 at this resolution; the companion paper reports what an independent active-fire observation says about the omission concern.

(iv) **Evia remains the most imbalance-atypical population** even
2070 after the AOI extension, at a TSG prevalence of 0.287 against 0.038 to 0.072 in three of the others.

(v) **Each region contributes one fire season**, so regional concept shift is confounded with event meteorology, and distinguishing them requires multi-year labels.

2075 (vi) **Cross-region point estimates carry an implementation tolerance** of roughly ± 0.02 to 0.03 across scikit-learn versions. All reported numbers are fixed to one verified version, but exact reproduction elsewhere requires the archived environment.

(vii) **The diagnostic correlations rest on an effective sample of**
2080 **ten region pairs.** Both the successes and the failures of Section 4.6 should be read at that power.

(viii) **Manavgat’s atypical transfer behaviour remains unexplained.**
It is where the conditional diagnosis bites hardest and where feature removal recovers most. Three candidates have now been tested and none survives:
2085 its meteorology, which was not extreme; the quality screening of its coarse thermal input, which propagates widely but moves no signed association by more than +0.0003 (Section 4.9, Appendix A(e)); and the evaluation frame, which explains its elevation figure but not its transfer behaviour (Section 4.4). With one fire season per region the remaining candidates are not separable
2090 in this design.

(ix) **The interval-support counts are less stable than the point estimates behind them.** Several verdicts sit within a thousandth of their reference value, and at 1 km blocking the published split of ten positive, seven negative and three uncertain turns on a lower bound of -0.00045 . The point
2095 estimates and the sign pattern are stable; the counts are not. Every sentence in this paper that leans on an exact count of supported directions should be read at that precision. (x) **The five areas of interest are not comparable frames, and this cohort cannot fully repair it** (Section 4.4). We report the equalised arm alongside the frame-as-drawn arm rather than replacing
2100 one with the other, because the collar radius is itself a choice and 5 km and 10 km do not agree exactly (0.608 against 0.617). The deeper limitation is that the frames were fixed upstream of this work, in **repo/**, so we can restrict them but not extend them; a region whose rectangle is already fire-scale, Montiferru, cannot be given a far field for symmetry. Any future cohort
2105 should fix the frame by an explicit accessible-area rule [3] before any predictor is computed, and we treat that as the main design lesson of this paper.

Appendix C.6. Leakage control

An explicit forbidden-column set is enforced at every model fit. Coordinates (**lon**, **lat**, **row**, **col** and their normalised forms), every burn-date and
2110 label-provenance column, and the agreement fraction are excluded from all feature sets, and the enforcement runs as an assertion rather than a convention. The natural-vegetation mask is used only to define the population, never as a predictor. Spatial blocking prevents a cell from sharing a fold with its own neighbours.

2115 **Reproducibility.** All randomness uses seed 42 and the bootstrap uses

1000 replicates throughout. The transfer and adaptation analysis runs in an environment separate from the upstream pipeline’s. Every within-region model was therefore refitted there and compared against the frozen upstream output, and the independently implemented adaptation was compared against the pipeline’s own. The within-region comparisons agree exactly and the twenty transfer directions to within 1.6×10^{-7} . All numbers here were produced under scikit-learn 1.9.0 or verified against it; the implementation tolerance that applies if the version is not fixed is stated in Section 5.9(vi), and the companion paper reports the version sensitivity and the reproduction check in full.

Sensitivity analyses. Every headline result is repeated across two analysis populations, three spatial-block sizes, the CORAL sweep, both feature sets and four classifier capacities; where a conclusion depends on one of those choices the dependence is reported rather than resolved by choosing the favourable setting (Appendix A).

Appendix C.7. Transfer-gap decomposition and the concept-shift diagnostic

For each direction the gap between the target’s own within-region skill and the raw transfer result is split in two. One part is what the best label-free adaptation recovers, and the other is what it does not. The recovered fraction is $(\text{adapted} - \text{raw}) / (\text{within} - \text{raw})$, signed and unclipped, with its interval from the same paired bootstrap; it bounds what covariate-level correction can achieve. Section 4.3 shows the remainder should not be read as a conditional residual, because much of it is incurred inside a single region (Appendix A(j)).

The mechanism is diagnosed by **signed univariate association**. For

each numeric predictor the raw ROC-AUC of that predictor against **burned** is computed in each region and never folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a value below 0.5 is read as a direction rather than as weakness. A reversal is called bootstrap-supported only when the two regions' point estimates fall on
2145 opposite sides of 0.5 **and each region's own interval excludes 0.5**, under the same 10-cell spatial-block bootstrap. That is stricter than requiring the two regions' intervals to be disjoint: a feature whose intervals are disjoint but one of which straddles 0.5 has not been shown to point anywhere in that region, so it is recorded as a point reversal only. Appendix B states
2150 the rule again beside the counts, and `conditional_similarity_transfer.json` carries it as machine-readable metadata.

CRedit authorship contribution statement

Yunus Emre Cogurcu: Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review and editing,
2155 Supervision. **Emrehan Metin:** Software, Data curation, Investigation, Validation, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work
2160 reported in this paper.

Funding

This work is an output of project 17506, supported by the Scientific Research Projects (BAP) unit of Çukurova University.

Data availability

2165 All satellite inputs are public and are obtained through Google Earth Engine. The processing pipeline, its configuration and the frozen numeric outputs are publicly available; the repository, the commit of record and the licence are given in the data and code availability statement of the Methods, together with the three components that fall outside that release. No digital
2170 object identifier is minted and no archival deposit exists.

References

- [1] Alkan Akıncı, H., Akıncı, H., 2023. Machine learning based forest fire susceptibility assessment of Manavgat district (Antalya), Turkey. *Earth Science Informatics* 16, 397–414. doi:10.1007/s12145-023-00953-5.
- 2175 [2] Archibald, S., Lehmann, C.E.R., Gómez-Dans, J.L., Bradstock, R.A., 2013. Defining pyromes and global syndromes of fire regimes. *Proceedings of the National Academy of Sciences* 110, 6442–6447. doi:10.1073/pnas.1211466110.
- [3] Barve, N., Barve, V., Jiménez-Valverde, A., Lira-Noriega, A., Maher, S.P., Peterson, A.T., Soberón, J., Villalobos, F., 2011. The
2180 crucial role of the accessible area in ecological niche modeling and

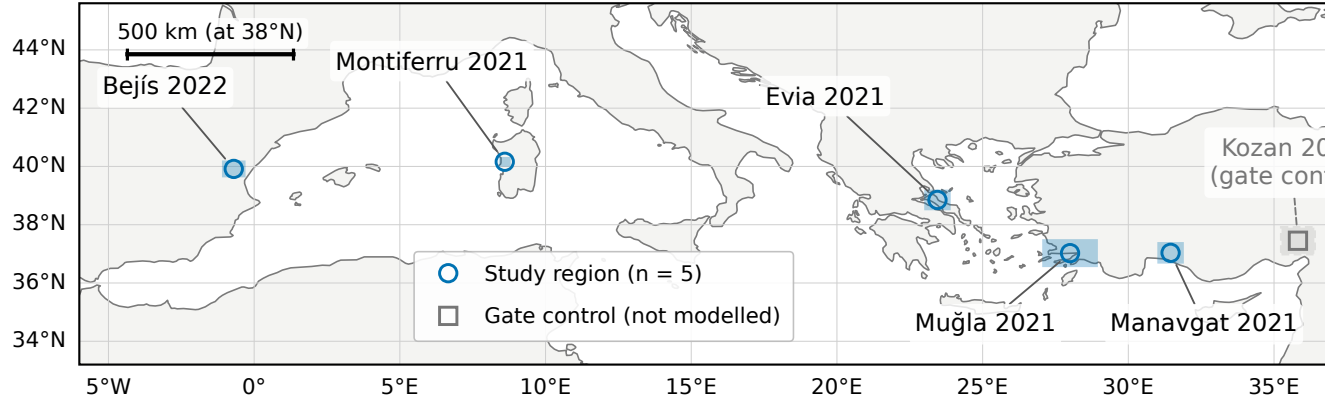


Figure C.1: **Study regions and the gate negative control.** Locations of the five wild-fire regions analysed in this study (blue, circular markers): Manavgat 2021 and Muğla 2021 (Türkiye), Bejís 2022 (Spain), North Evia 2021 (Greece, extended AOI) and Montiferru 2021 (Sardinia, Italy). Rectangles are the true analysis areas of interest; the ring marks each centroid, because at basin scale the rectangles alone are hard to locate. Kozan 2023 (grey, dashed outline, square marker) is *not* a study region: it is the negative control used to calibrate the burned-landcover admissibility gate, and it enters no model, transfer or diagnostic anywhere in this paper. Its rectangle is the true extent of its processed AOI and is larger than the study AOIs for that reason alone; the difference in size carries no methodological meaning and is not a design choice. Coastlines are Natural Earth 1:50 m; country borders are omitted deliberately. The scale bar is exact only at 38°N. Region bounding boxes match Table 1 to 5×10^{-3} degrees.

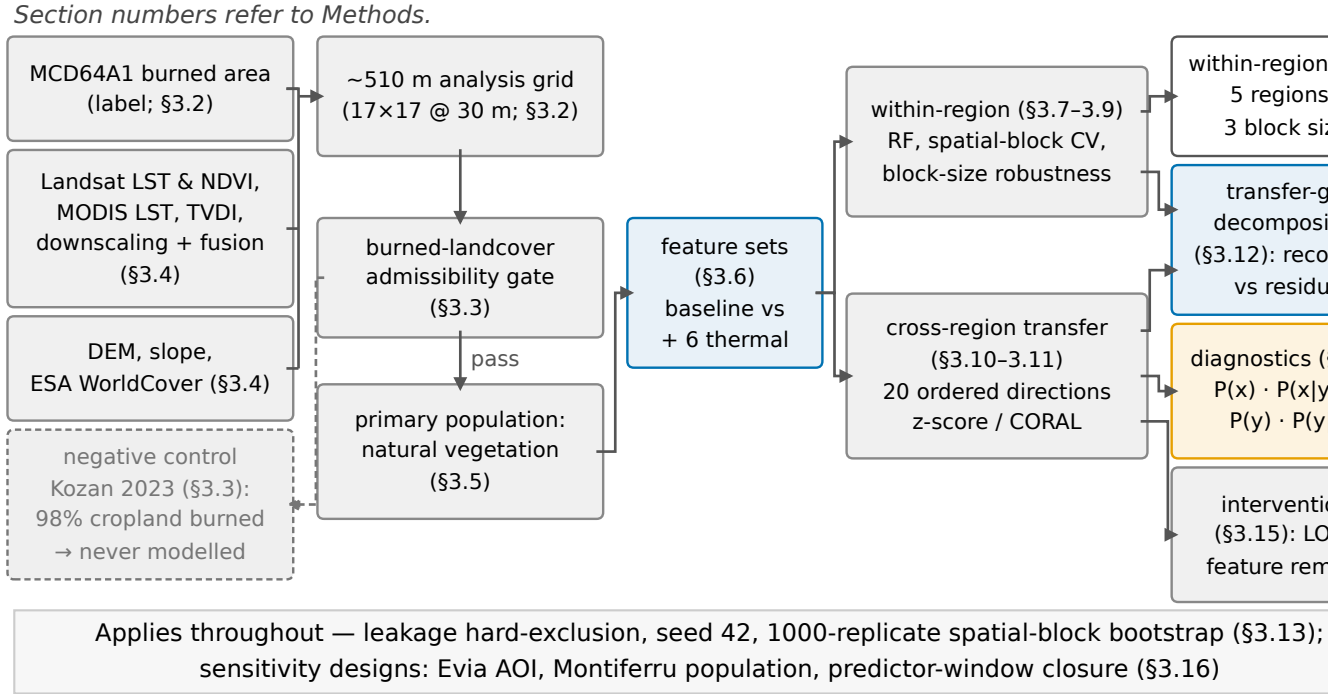


Figure C.2: Processing pipeline and evaluation programme. Flow from source products (left) to the transfer and diagnostic programme (right); every box carries the Methods subsection that specifies it. The five source products (MCD64A1 burned area, Landsat Collection 2 Level-2, MOD11A1, Copernicus DEM GLO-30 and ESA WorldCover) merge onto the reconstructed ~ 510 m analysis grid (17×17 pixels of the 30 m reference grid). The burned-landcover admissibility gate has two outcomes: regions that *pass* proceed to the natural-vegetation primary population, while the rejecting branch (grey, dashed) terminates in the Kozan 2023 negative control, whose burned area is 98% cropland and which is never modelled. The dashed connector deliberately carries no arrowhead onward, so the control cannot be read as entering the pipeline. Feature sets feed two evaluations in parallel: within-region (random forest under spatial-block cross-validation) and cross-region transfer (source-only, 20 ordered directions, with label-blind z-score and CORAL adaptation). Their levels are combined in the transfer-gap decomposition (§3.12). The footer states what applies throughout: leakage hard-exclusion, seed 42, 1000-replicate spatial-block bootstrap (§3.13), and the sensitivity designs (§3.16). Section references are verified against the Methods heading list when the figure is built.

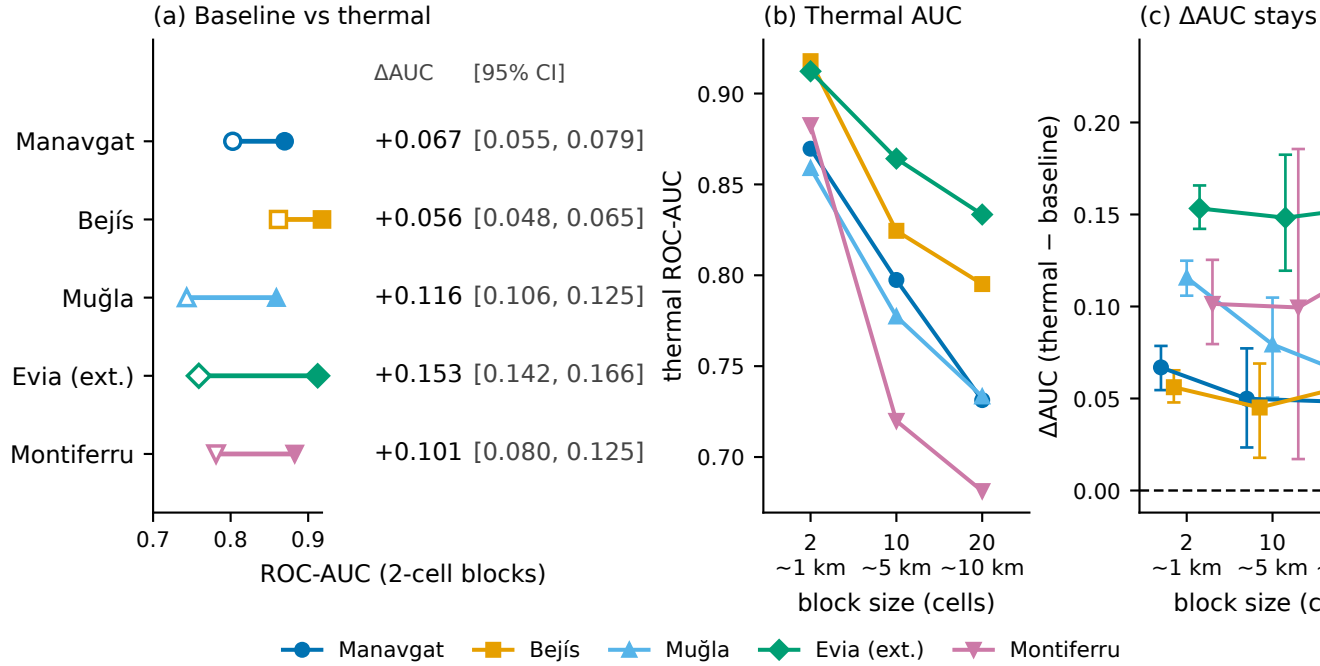


Figure C.3: **The thermal increment replicates in all five regions and survives coarser spatial blocking.** Natural-vegetation primary population; random forest with the primary weighted configuration; all intervals are 95% spatial-block bootstrap intervals (1000 replicates, seed 42). **(a)** Baseline versus thermal out-of-fold ROC-AUC per region at the default 2-cell (~ 1 km) blocking: the open marker is the static baseline feature set, the filled marker is the baseline plus the six thermal predictors, and the connecting bar is the increment. The right-hand column reports the paired ΔAUC and its 95% interval; it is a fixed column placed beyond the plotted AUC range, so no label position depends on a data value. Every interval excludes zero. **(b)** Absolute thermal ROC-AUC against spatial block size (2, 10 and 20 cells, ≈ 1 , 5 and 10 km): absolute skill falls as blocking coarsens, in every region. **(c)** The thermal increment ΔAUC against the same block sizes, with 95% intervals: the increment does *not* melt away, and its interval excludes zero in all ten of the 2- and 10-cell region-block-size combinations. The five 20-cell intervals also lie above zero but rest on 6 to 33 positive-carrying blocks each, so they are plotted as indicative rather than as intervals (Table 3, table note). In (c) the five regions are offset horizontally by 0.15 cell so their whiskers separate; **this offset is cosmetic, since all five regions are evaluated at exactly the same three block sizes** (2, 10, 20 cells), and the offset is smaller than half the spacing between block sizes, so the groups cannot merge. Colour and marker shape both encode region, so the panels remain readable in greyscale.

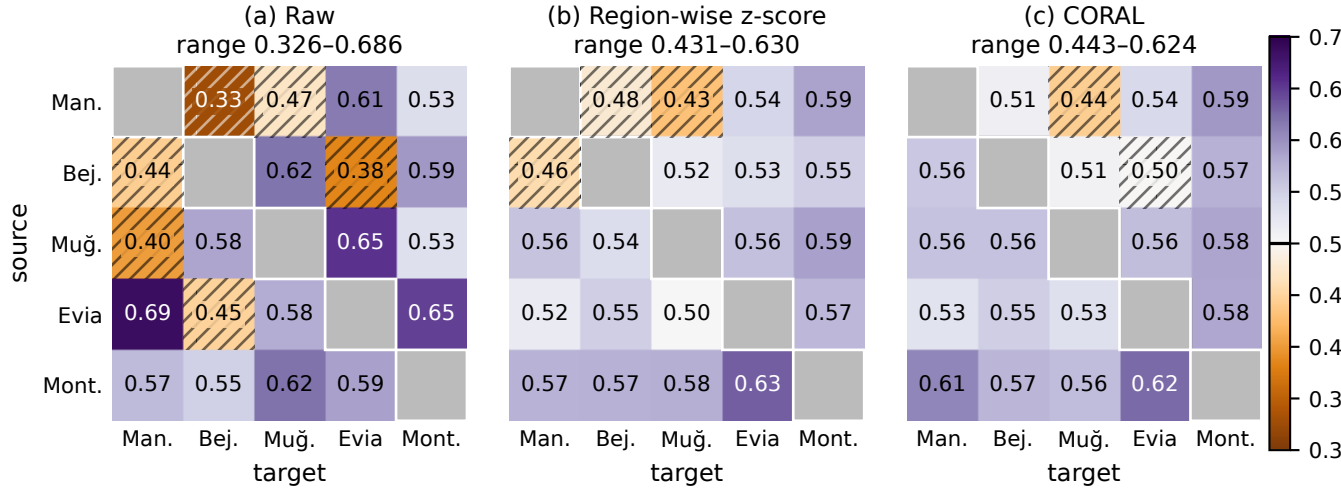


Figure C.4: **Cross-region transfer, and what label-blind adaptation does to it.** Thermal transfer ROC-AUC for all 20 ordered source–target directions among the five regions, natural-vegetation primary population. (a) Raw transfer, no adaptation. (b) After region-wise z-score standardisation. (c) After CORAL. Rows are the source region, columns the target; the grey diagonal is within-region performance and is *not* a transfer result. The diverging colour scale is centred exactly on chance (0.5, marked by the black rule on the colour bar), so orange is below chance and purple above. Each panel header states that panel’s range: adaptation compresses the spread from 0.326–0.686 (raw) to 0.431 to 0.630 (z-score) and 0.443 to 0.624 (CORAL). It pulls the extremes toward chance rather than lifting transfer. Cells below chance are hatched. The hatch is a redundant cue for print: a diverging palette is symmetric in luminance about its centre, so in greyscale a below-chance cell would otherwise be indistinguishable from an equally distant above-chance one. Hatching follows the *exact* value while the printed label is rounded to two decimals, so two cells print “0.50” with different hatching: CORAL Bejís→Evia is 0.4991 (hatched, below chance) and z-score Evia→Muğla is 0.5013 (un-hatched, above chance). Every cell carries its value, and every label’s colour is chosen by measured WCAG contrast against that cell’s rendered colour (worst case 4.7:1). Confidence intervals for these directions are given in Table 4.

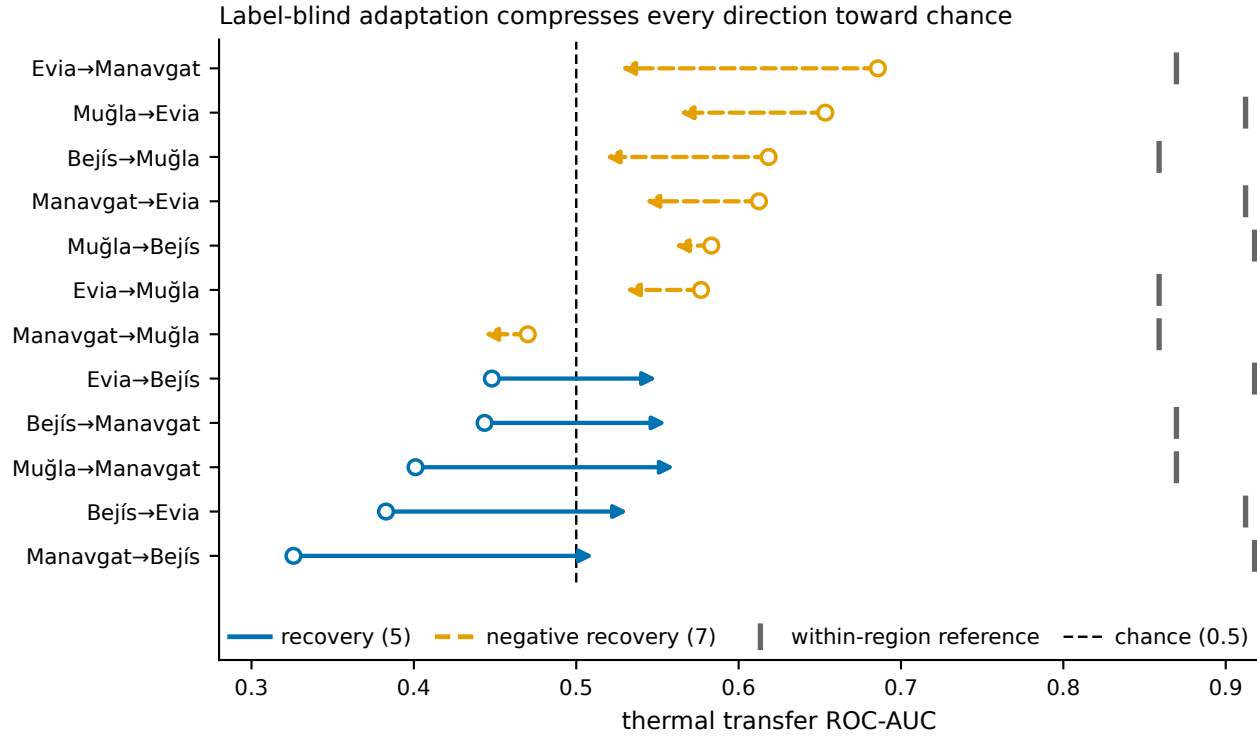


Figure C.5: **Label-blind adaptation regresses transfer onto the level a model reaches on an unseen scar, rather than recovering it.** All 12 ordered directions among the four regions carrying the full decomposition (Manavgat, Bejís, Muğla, Evia), natural-vegetation primary population. Each row is one direction. The open circle is raw transfer ROC-AUC and the arrow runs to the best label-blind adapted result for that direction (region-wise z-score or CORAL, whichever is better); the grey vertical tick is the within-region reference for that target, and the dashed vertical rule is chance. **Solid blue arrows are the five directions where adaptation moves transfer toward the within-region reference (recovery); dashed orange arrows are the seven where it moves *away* from it (negative recovery).** Line style and colour both encode this class, because the two hues differ by only 2.30:1 in relative luminance and would nearly merge in greyscale. The pattern is the point: all six directions starting above chance are pushed down, and five of the six starting below chance are pushed up, so adaptation acts on the spread rather than on skill. The exception is Manavgat→Muğla, which starts below chance at 0.470 and is pushed further down to 0.443. The worst case is Evia→Manavgat, where raw transfer of 0.686 is dragged down to 0.527 against a within-region reference of 0.870, a recovered fraction of -0.862 . No arrow reaches its within-region tick.

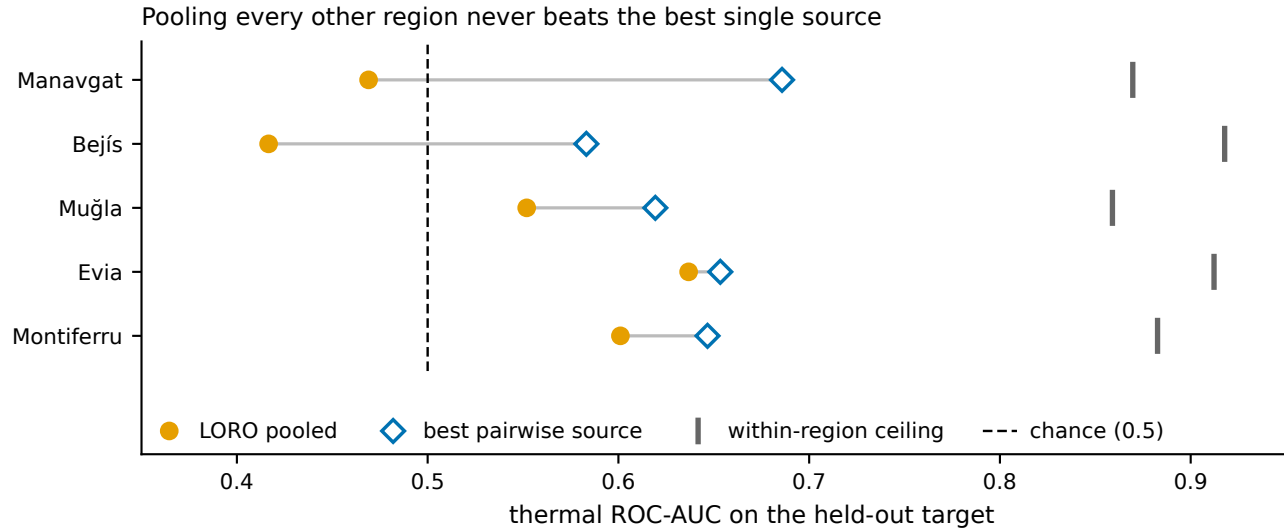


Figure C.6: **Pooling every other region never beats the best single source.** Leave-one-region-out evaluation: for each held-out target region (rows), a model trained on the pooled remaining four regions (filled orange circle) is compared against the best individual source region for that same target (open blue diamond), with the within-region ceiling as a grey vertical tick and chance as the dashed rule. Natural-vegetation primary population, thermal feature set, no adaptation. The pooled model lies to the left of the best pairwise source in all five targets without exception. Adding source regions does not accumulate transferable signal. In two targets the pooled model falls *below chance* at the point estimate (Manavgat 0.469, Bejís 0.417), and only for Bejís does the interval lie entirely below chance, while a single well-chosen source reaches 0.686 and 0.583 respectively. Every point remains far below the within-region ceiling.

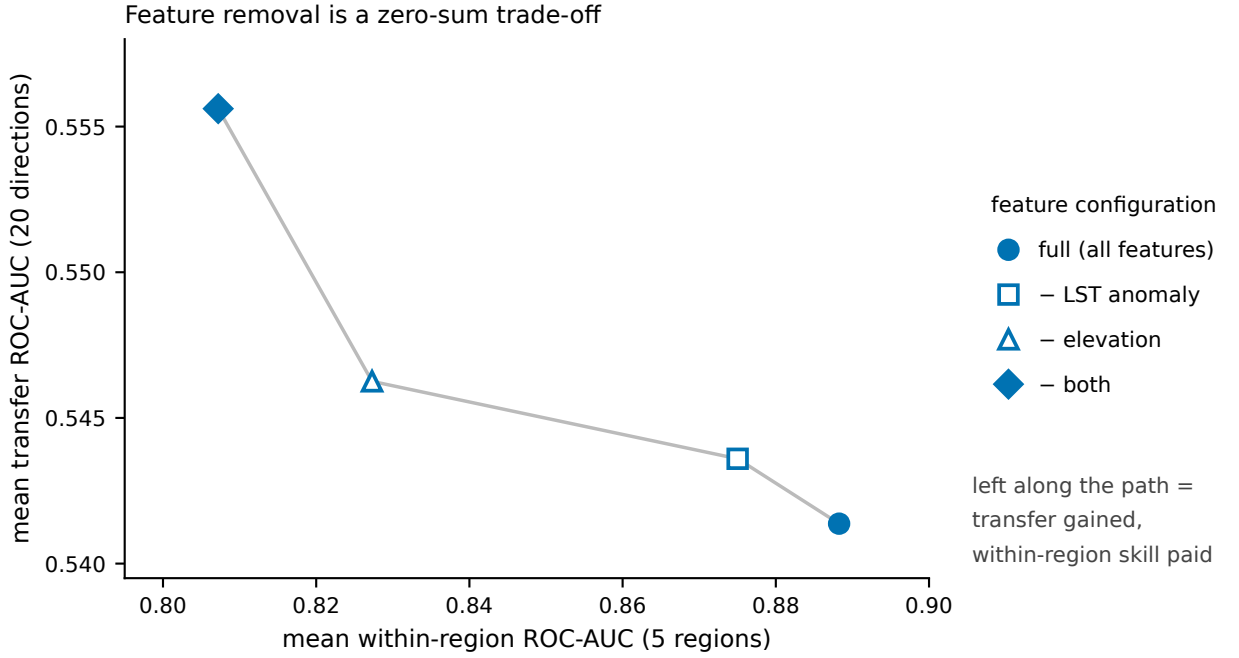


Figure C.7: **Removing the direction-reversing features costs locally and returns nothing measurable on transfer.** Mean within-region ROC-AUC across the five regions (horizontal) against mean transfer ROC-AUC across the 20 ordered directions (vertical), for four feature configurations: the full thermal set, the set without the LST anomaly, without elevation, and without both. Each configuration has its own marker and is named in the key; the grey path connects them in that order. Moving left along the path, transfer rises monotonically ($0.541 \rightarrow 0.556$) while within-region skill falls monotonically ($0.888 \rightarrow 0.807$). Both monotonicities are verified when the figure is built, so the figure cannot silently contradict the claim. The within-region cost is 0.081 AUC and is supported in every region. The change in transfer is +0.014 AUC on an interval that spans zero, so no exchange between the two is demonstrated: a local cost is measured and no compensating transfer gain is.

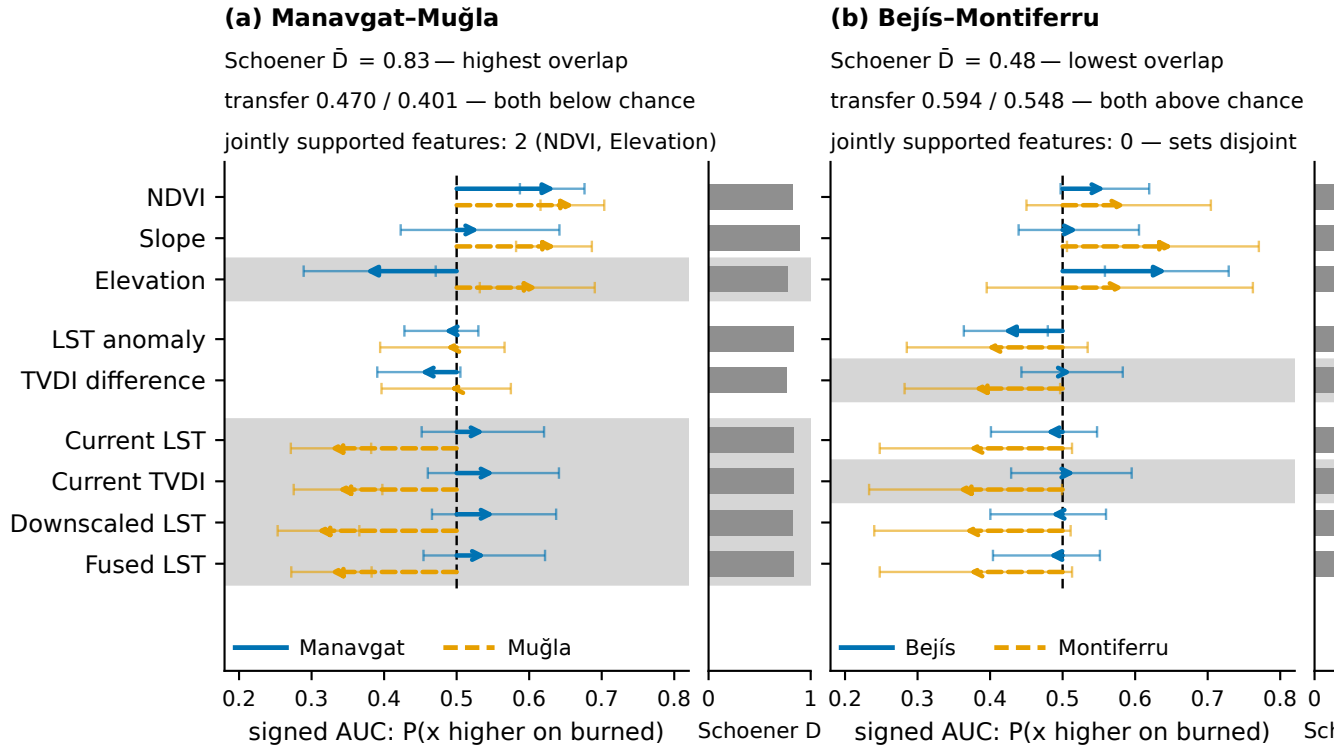


Figure C.8: **Burned-niche overlap does not determine transfer: the contrast pair.** Per-feature signed univariate AUC, the probability that a predictor takes a higher value on a burned cell than on an unburned one, for the two most informative region pairs, natural-vegetation primary population. Arrows run from chance (0.5) to the observed value, so their direction is the sign of the association and their length its strength; whiskers are 95% spatial-block (≈ 5 km) bootstrap intervals. Within each pair the two regions are distinguished by colour, by line style (solid for the first region named, dashed for the second) and by vertical offset; line style is included because the two hues differ by only 2.30:1 in relative luminance and would nearly merge in greyscale. **A filled arrowhead means that region's 95% interval excludes 0.5, so the direction is statistically supported; an open arrowhead means the interval contains 0.5 and the direction is not supported.** Rows shaded grey are features whose sign disagrees between the two regions. Grey side bars are the per-feature Schoener's D overlap. **(a)** Manavgat–Muğla, the pair with the *highest* burned-niche overlap ($\bar{D} = 0.83$), transfers below chance in both directions (0.470 and 0.401). **(b)** Bejís–Montiferru, the pair with the *lowest* overlap ($\bar{D} = 0.48$), transfers above chance in both directions (0.594 and 0.548). High niche overlap is therefore not sufficient for transfer. The arrowhead fill also exposes the coverage limit of the supported-agreement index, which is defined only on features whose direction is supported in *both* regions: there are two such features in (a) (NDVI and elevation) and **none in (b)**, where the two regions' supported sets are disjoint and Montiferru's intervals

species distribution modeling. *Ecological Modelling* 222, 1810–1819. doi:10.1016/j.ecolmodel.2011.02.011.

- 2185 [4] Boschetti, L., Roy, D.P., Giglio, L., Huang, H., Zubkova, M., Humber, M.L., 2019. Global validation of the Collection 6 MODIS burned area product. *Remote Sensing of Environment* 235, 111490. doi:10.1016/j.rse.2019.111490.
- [5] Breiman, L., 2001. Random forests. *Machine Learning* 45, 5–32. doi:10.1023/A:1010933404324.
- 2190 [6] de Bruin, S., Brus, D.J., Heuvelink, G.B.M., van Ebbenhorst Tengbergen, T., Wadoux, A.M.J.C., 2022. Dealing with clustered samples for assessing map accuracy by cross-validation. *Ecological Informatics* 69, 101665. doi:10.1016/j.ecoinf.2022.101665.
- 2195 [7] Chuvieco, E., Cocero, D., Riaño, D., Martín, P., Martínez-Vega, J., de la Riva, J., Pérez, F., 2004. Combining NDVI and surface temperature for the estimation of live fuel moisture content in forest fire danger rating. *Remote Sensing of Environment* 92, 322–331. doi:10.1016/j.rse.2004.01.019.
- 2200 [8] Dimarco, N.A., Faraji, I., Wahbi, M., Maatouk, M., Boulaassal, H., Yazidi Aalaoui, O., El Kharki, O., 2026. Generalizing human-driven wildfire ignition models across Mediterranean regions using harmonized remote-sensing and machine-learning data. *Geomatics* 6, 13. doi:10.3390/geomatics6010013.

- 2205 [9] Gelabert, P.J., Jiménez-Ruano, A., Ochoa, C., Alcasena, F., Sjöström, J., Marrs, C., Ribeiro, L.M., Palaiologou, P., Bentu Martínez, C., Chuvieco, E., Vega-García, C., Rodrigues, M., 2025. Assessing human-caused wildfire ignition likelihood across Europe. *Natural Hazards and Earth System Sciences* 25, 4713–4729. doi:10.5194/nhess-25-4713-2025.
- 2210 [10] Giglio, L., Boschetti, L., Roy, D.P., Humber, M.L., Justice, C.O., 2018. The Collection 6 MODIS burned area mapping algorithm and product. *Remote Sensing of Environment* 217, 72–85. doi:10.1016/j.rse.2018.08.005.
- 2215 [11] Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., Moore, R., 2017. Google Earth Engine: Planetary-scale geospatial analysis for everyone. *Remote Sensing of Environment* 202, 18–27. doi:10.1016/j.rse.2017.06.031.
- 2220 [12] Huang, Y., Liu, N., Chen, W., Tan, W., Deng, Y., Wang, Z., Chlus, A., Shen, J., Townsend, P.A., 2026. Enhancing transferability of foliar trait retrieval models: A comparative analysis of transfer learning strategies and domain shift characterization. *Remote Sensing of Environment* 337, 115336. doi:10.1016/j.rse.2026.115336.
- 2225 [13] Iban, M.C., Sekertekin, A., 2022. Machine learning based wildfire susceptibility mapping using remotely sensed fire data and GIS: A case study of Adana and Mersin provinces, Turkey. *Ecological Informatics* 69, 101647. doi:10.1016/j.ecoinf.2022.101647.

- [14] Jodhani, K., Bhatiya, L., Sirvi, L., Chanda, A., Thakkar, S., Rathnayake, U., 2026. Artificial intelligence and remote sensing frameworks for wildfire monitoring and risk analysis across multiple ecosystems: A review. *AIMS Environmental Science* 13, 56–98. doi:10.3934/environsci.2026004.
- [15] Kondylatos, S., Prapas, I., Camps-Valls, G., Papoutsis, I., 2023. Meso-geos: A multi-purpose dataset for data-driven wildfire modeling in the Mediterranean, in: *Advances in Neural Information Processing Systems* 36 (NeurIPS 2023) Datasets and Benchmarks Track, pp. 50661–50676. doi:10.52202/075280–2204.
- [16] Liu, C., Mostafavi, A., 2025. WildfireGenome: Interpretable machine learning reveals local drivers of wildfire risk and their cross-county variation. *arXiv preprint arXiv:2511.11589*. doi:10.48550/arXiv.2511.11589, arXiv:2511.11589.
- [17] Lobo, J.M., Jiménez-Valverde, A., Real, R., 2008. AUC: a misleading measure of the performance of predictive distribution models. *Global Ecology and Biogeography* 17, 145–151. doi:10.1111/j.1466-8238.2007.00358.x.
- [18] Ludwig, M., Moreno-Martinez, A., Hölzel, N., Pebesma, E., Meyer, H., 2023. Assessing and improving the transferability of current global spatial prediction models. *Global Ecology and Biogeography* 32, 356–368. doi:10.1111/geb.13635.
- [19] Maffei, C., Alfieri, S.M., Menenti, M., 2018. Relating spatiotemporal

- 2250 patterns of forest fires burned area and duration to diurnal land surface temperature anomalies. *Remote Sensing* 10, 1777. doi:10.3390/rs10111777.
- [20] Maffei, C., Lindenbergh, R., Menenti, M., 2021. Combining multispectral and thermal remote sensing to predict forest fire characteristics. *ISPRS Journal of Photogrammetry and Remote Sensing* 181, 400–412. 2255 doi:10.1016/j.isprsjprs.2021.09.016.
- [21] Maffei, C., Menenti, M., 2019. Predicting forest fires burned area and rate of spread from pre-fire multispectral satellite measurements. *ISPRS Journal of Photogrammetry and Remote Sensing* 158, 263–278. doi:10. 2260 1016/j.isprsjprs.2019.10.013.
- [22] Marino, E., Yáñez, L., Guijarro, M., Madrigal, J., Senra, F., Rodríguez, S., Tomé, J.L., 2024. Transferability of empirical models derived from satellite imagery for live fuel moisture content estimation and fire risk prediction. *Fire* 7, 276. doi:10.3390/fire7080276.
- 2265 [23] Meyer, H., Pebesma, E., 2021. Predicting into unknown space? Estimating the area of applicability of spatial prediction models. *Methods in Ecology and Evolution* 12, 1620–1633. doi:10.1111/2041-210X.13650.
- [24] Meyer, H., Pebesma, E., 2022. Machine learning-based global maps of ecological variables and the challenge of assessing them. *Nature Com-* 2270 *munications* 13, 2208. doi:10.1038/s41467-022-29838-9.
- [25] Meyer, H., Reudenbach, C., Hengl, T., Katurji, M., Nauss, T., 2018. Improving performance of spatio-temporal machine learning models using

- forward feature selection and target-oriented validation. *Environmental Modelling & Software* 101, 1–9. doi:10.1016/j.envsoft.2017.12.001.
- 2275 [26] Milà, C., Mateu, J., Pebesma, E., Meyer, H., 2022. Nearest neighbour distance matching leave-one-out cross-validation for map validation. *Methods in Ecology and Evolution* 13, 1304–1316. doi:10.1111/2041-210X.13851.
- [27] Moreno-Torres, J.G., Raeder, T., Alaiz-Rodríguez, R., Chawla, N.V.,
2280 Herrera, F., 2012. A unifying view on dataset shift in classification. *Pattern Recognition* 45, 521–530. doi:10.1016/j.patcog.2011.06.019.
- [28] Muñoz-Sabater, J., Dutra, E., Agustí-Panareda, A., Albergel, C., Arduini, G., Balsamo, G., Boussetta, S., Choulga, M., Harrigan, S., Hersbach, H., Martens, B., Miralles, D.G., Piles, M., Rodríguez-Fernández, N.J., Zsoter, E., Buontempo, C., Thépaut, J.N., 2021. ERA5-Land: a state-of-the-art global reanalysis dataset for land applications. *Earth System Science Data* 13, 4349–4383. doi:10.5194/essd-13-4349-2021.
2285
- [29] Pausas, J.G., Keeley, J.E., 2021. Wildfires and global change. *Frontiers in Ecology and the Environment* 19, 387–395. doi:10.1002/fee.2359.
- 2290 [30] Persello, C., Bruzzone, L., 2012. Active learning for domain adaptation in the supervised classification of remote sensing images. *IEEE Transactions on Geoscience and Remote Sensing* 50, 4468–4483. doi:10.1109/TGRS.2012.2192740.
- [31] Ploton, P., Mortier, F., Réjou-Méchain, M., Barbier, N., Picard, N.,
2295 Rossi, V., Dormann, C., Cornu, G., Viennois, G., Bayol, N., Lyapustin,

- A., Gourlet-Fleury, S., Pélissier, R., 2020. Spatial validation reveals poor predictive performance of large-scale ecological mapping models. *Nature Communications* 11, 4540. doi:10.1038/s41467-020-18321-y.
- [32] Podschwit, H., Jolly, W., Alvarado, E., Verma, S., Ponce, B., Markos, A., Aliaga-Nestares, V., Rodriguez-Zimmermann, D., 2022. Reliability of cross-regional applications of global fire danger models: A Peruvian case study. *Fire Ecology* 18, 25. doi:10.1186/s42408-022-00150-7.
- [33] Roberts, D.R., Bahn, V., Ciuti, S., Boyce, M.S., Elith, J., Guillera-Aroita, G., Hauenstein, S., Lahoz-Monfort, J.J., Schröder, B., Thuiller, W., Warton, D.I., Wintle, B.A., Hartig, F., Dormann, C.F., 2017. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography* 40, 913–929. doi:10.1111/ecog.02881.
- [34] Rousseau, J.S., Betts, M.G., 2022. Factors influencing transferability in species distribution models. *Ecography* 2022, e06060. doi:10.1111/ecog.06060.
- [35] Sandholt, I., Rasmussen, K., Andersen, J., 2002. A simple interpretation of the surface temperature/vegetation index space for assessment of surface moisture status. *Remote Sensing of Environment* 79, 213–224. doi:10.1016/S0034-4257(01)00274-7.
- [36] Schoener, T.W., 1968. The *Anolis* lizards of Bimini: Resource partitioning in a complex fauna. *Ecology* 49, 704–726. doi:10.2307/1935534.

- [37] Sun, B., Feng, J., Saenko, K., 2016. Return of frustratingly easy domain adaptation, in: Proceedings of the Thirtieth AAAI Conference on Artificial Intelligence (AAAI-16), AAAI Press, Phoenix, Arizona, USA. pp. 2058–2065. doi:10.1609/aaai.v30i1.10306.
- [38] Tuia, D., Persello, C., Bruzzone, L., 2016. Domain adaptation for the classification of remote sensing data: An overview of recent advances. IEEE Geoscience and Remote Sensing Magazine 4, 41–57. doi:10.1109/MGRS.2016.2548504.
- [39] Valavi, R., Elith, J., Lahoz-Monfort, J.J., Guillera-Arroita, G., 2019. blockCV: An R package for generating spatially or environmentally separated folds for k-fold cross-validation of species distribution models. Methods in Ecology and Evolution 10, 225–232. doi:10.1111/2041-210X.13107.
- [40] VanDerWal, J., Shoo, L.P., Graham, C., Williams, S.E., 2009. Selecting pseudo-absence data for presence-only distribution modeling: How far should you stray from what you know? Ecological Modelling 220, 589–594. doi:10.1016/j.ecolmodel.2008.11.010.
- [41] Vesk, P.A., Morris, W.K., Neal, W.C., Mokany, K., Pollock, L.J., 2021. Transferability of trait-based species distribution models. Ecography 44, 134–147. doi:10.1111/ecog.05179.
- [42] Vibhandik, P., Sawant, S., Joshi, A., Bidwe, R., 2026. A systematic literature review on forest fire susceptibility mapping using geo-spatial

- 2340 technology and future research directions. *Discover Artificial Intelligence* 6, 921. doi:10.1007/s44163-026-00921-0.
- [43] Wadoux, A.M.J.C., Heuvelink, G.B.M., de Bruin, S., Brus, D.J., 2021. Spatial cross-validation is not the right way to evaluate map accuracy. *Ecological Modelling* 457, 109692. doi:10.1016/j.ecolmodel.2021.109692.
- 2345 [44] Warren, D.L., Glor, R.E., Turelli, M., 2008. Environmental niche equivalency versus conservatism: Quantitative approaches to niche evolution. *Evolution* 62, 2868–2883. doi:10.1111/j.1558-5646.2008.00482.x.
- [45] Xu, Y., Dai, Y., Chang, L., Wang, Q., Dong, Y., 2026. Does your wildfire prediction model actually work, or just score well? arXiv preprint arXiv:2605.18911. doi:10.48550/arXiv.2605.18911, arXiv:2605.18911.
- 2350 [46] Yates, K.L., Bouchet, P.J., Caley, M.J., Mengersen, K., Randin, C.F., Parnell, S., et al., 2018. Outstanding challenges in the transferability of ecological models. *Trends in Ecology & Evolution* 33, 790–802. doi:10.1016/j.tree.2018.08.001.
- 2355 [47] Yebra, M., Dennison, P.E., Chuvieco, E., Riaño, D., Zylstra, P., Hunt, E.R., Danson, F.M., Qi, Y., Jurdao, S., 2013. A global review of remote sensing of live fuel moisture content for fire danger assessment: Moving towards operational products. *Remote Sensing of Environment* 136, 455–468. doi:10.1016/j.rse.2013.05.029.
- 2360

- [48] Zanaga, D., Van De Kerchove, R., Daems, D., De Keersmaecker, W., Brockmann, C., Kirches, G., Wevers, J., Cartus, O., Santoro, M., Fritz, S., Lesiv, M., Herold, M., Tsendbazar, N.E., Xu, P., Ramoino, F., Arino, O., 2022. ESA WorldCover 10 m 2021 v200. Zenodo. doi:10.5281/zenodo.7254221.